

Equilibrium from the inside out: The relativistic structure of Quantum Thermodynamics

Gerard McCaul

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We obtain the exact Hamiltonian of mean force for a finite-dimensional quantum system at arbitrary coupling to a Gaussian environment - non-perturbative, closed-form, and valid at all temperatures. The key is to trace the bath *before* expanding: exactness then follows from the boundedness of the system algebra, not from resumming a divergent series. For a qubit with non-commuting coupling, the solution exposes a hidden geometric structure. The bath breaks the orientability of the influence state, forcing the reduced equilibrium into one of two dual representations related by an involutive map. In natural thermal coordinates this map is a Lorentz boost: a fixed sum of rapidities, rectangular hyperbolas, a light cone, a Minkowski metric. The crossover between weak and strong coupling is governed by a single universal function that simultaneously maximises the extractable quantum work, and the ultrastrong limit of Cresser and Anders emerges as the zero-coupling asymptotic of the dual frame. For N -level systems the metric generalises: the spectrum of transition frequencies couples to the physical temperature through a renormalisation tensor, and the passage from special to general relativity is immediate.

INTRODUCTION

What does equilibrium look like from the inside? In closed quantum statistical mechanics the answer is immediate: the thermal state is generated by a Hamiltonian, $\rho \propto e^{-\beta H}$. For a subsystem at finite coupling to its environment, however, the operationally defined equilibrium state is not $e^{-\beta H_Q}$ for any bare Hamiltonian H_Q . It is, instead, the partial trace of the global Gibbs state,

$$\bar{\rho}_Q(\beta) = \text{Tr}_X e^{-\beta H_{\text{tot}}}, \quad (1)$$

and the object whose Gibbs form reproduces this state - the Hamiltonian of mean force (HMF) - is a temperature-dependent, interaction-dressed operator that encodes the full non-perturbative influence of the environment on the subsystem [1–5]. The HMF is not an optional reinterpretation of the reduced state; it is its exact generator,

$$e^{-\beta H_{\text{MF}}(\beta)} \propto \frac{\text{Tr}_X e^{-\beta H_{\text{tot}}}}{Z_X(\beta)}, \quad Z_X(\beta) = \text{Tr}_X e^{-\beta H_X}. \quad (2)$$

Every quantum device, every biological subsystem, every qubit in a processor exists at finite coupling. When the coupling is non-negligible, the HMF is the sole object that correctly initialises correlations, correctly defines heat and work, and correctly furnishes the free-energy and fluctuation relations on which strong-coupling thermodynamics is built [6–15].

The problem is not whether this object exists - it is defined by the logarithm of a traced operator and is therefore always available in principle. The problem is whether it can be *written down*. Existing approaches fall into three categories, none of which resolves the question in general. In solvable limits - commuting (QND) interactions, purely quadratic models, singular ultrastrong projections - the operator algebra closes and compact forms are available [16–24]. Perturbative methods - weak-

coupling expansions, reaction-coordinate mappings, polaron transforms - produce controlled series that break down at the coupling strengths where the physics becomes most interesting [25–31]. Numerical methods - imaginary-time path integrals, hierarchical equations of motion, stochastic Liouville approaches - can compute $\bar{\rho}_Q(\beta)$ to arbitrary accuracy but do not yield an operator form [32–39]. The fundamental obstruction is always the same: conventional expansions perturb in a coupling that involves the unbounded bath operator, and the resulting series has no natural truncation at strong coupling.

This paper presents a route that avoids the obstruction entirely. Building on the quenched-density construction of Ref. [40], we trace the bath *before* performing any expansion. For a Gaussian (harmonic) environment this is exact: the cumulant theorem resums every order of the bath noise at every coupling strength, and the surviving influence operator $\Delta(\beta)$ is a bounded operator on the finite-dimensional system Hilbert space [17, 41–44]. The subsequent Baker–Campbell–Hausdorff step [45–47] is an algebraic operation on bounded operators, and for any finite-dimensional system the adjoint chain of the coupling operator terminates by the Cayley–Hamilton theorem. The resulting mean-force Hamiltonian is exact - non-perturbative in the coupling strength, valid at all temperatures, and expressed in closed form within the natural operator algebra of the system.

The consequences of this are remarkably far-reaching, and uniquely quantum mechanical. In the simplest non-trivial instance - a qubit interacting with a harmonic bath via a coupling that does not commute with the qubit Hamiltonian - the exact solution reveals a geometric organisation of the Bloch sphere that has no analogue in the weak-coupling or commuting limits. The influence of the bath is encoded in a symmetrised state with a discrete orientability symmetry. This symmetry is broken when the influence is recombined with the bare Gibbs factor

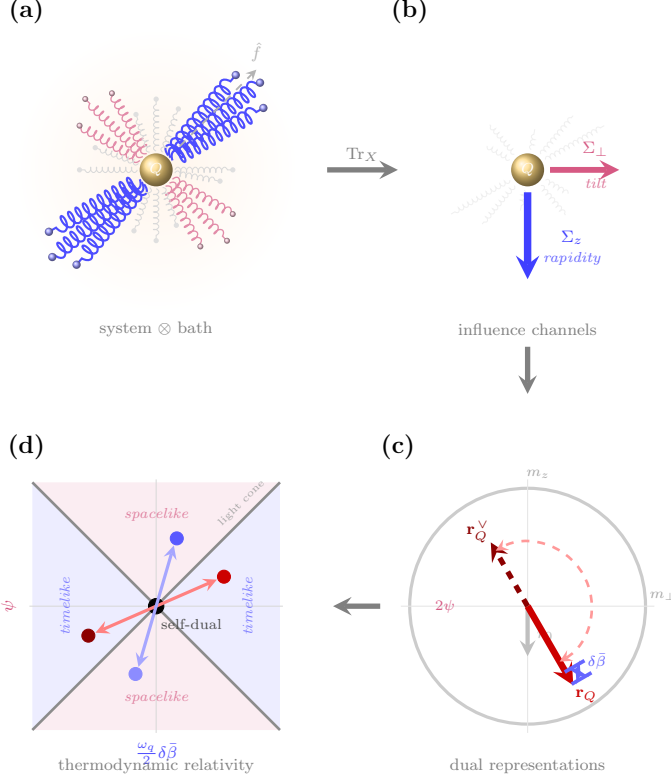


FIG. 1. From anisotropic bath influence to thermodynamic relativity. (a) A qubit coupled to a harmonic bath; the coupling $\hat{f}$ (dashed) makes the influence anisotropic. (b) After the partial trace, the environment collapses to two channels: Σ_z (blue, rapidity/temperature renormalisation) and $\Sigma_\perp$ (purple, tilt). (c) The resulting mean-force state $\mathbf{r}_Q$ (solid) and its dual $\mathbf{r}_Q^V$ (dashed), with the radial separation $\delta\bar{\beta}$ (blue) and angular gap 2ψ (purple) shown explicitly. (d) These separations define the axes of a Minkowski metric in thermal coordinates. Temperature renormalisation $\rightarrow$ timelike; tilt $\rightarrow$ spacelike. The dual map is point reflection through the self-dual origin ($\bullet$).

to form the reduced state. The broken symmetry forces a dual representation: two complementary parametrisations of the same physical state, related by an involutive map. Written in natural thermal coordinates, this map turns out to be a fixed sum of rapidities - the kinematics of a Lorentz boost. The dual curves are rectangular hyperbolas; the self-dual fixed point sits at the vertex of a light cone; and the full dual structure carries the signature of a Minkowski metric on the space of renormalised temperatures and tilt angles (Fig. 1). For a two-level system, the single transition frequency plays the role of mass, and the duality recovers the kinematics of special relativity. For an N -level system, the spectrum of transition frequencies couples to the physical temperature through a thermal renormalisation tensor, and the metric acquires the structure of a field equation: the passage from special to general relativity.

But the geometry is not merely elegant. The exact so-

lution furnishes a two-dimensional running coupling and a universal crossover function that interpolates between the weak- and strong-coupling regimes. The crossover region is precisely where the state is most sensitive to the bath - and precisely where extractable work is maximised: the coherence-enabled ergotropy of a qubit engine peaks at the crossover, establishing a structural quantum advantage valid for any positive spectral density. In the ultrastrong limit, the dual chart reproduces the coupling-basis thermal state of Cresser and Anders [20, 48] - not as a separate limiting calculation, but as the asymptotic value of the exact dual flow. The same involutive mapping that organises the geometry also promises a practical tool: quantities inaccessible in one representation become elementary in the other.

The development proceeds as follows. Section recapitulates the quenched representation. Section establishes the mean-force Hamiltonian and the Gaussian specialisation. Section derives the generating-function structure of the influence operator and identifies precise closure conditions. Section instantiates these to yield an exact HMF for a Pauli-closed qubit. The geometric structure of the resulting mean-force state and the dual representations are developed in Sec. . Section explores the physical regimes - purity bounds, running couplings, crossover physics, quantum engine advantage, and the ultrastrong duality. The paper closes with a Discussion (Sec.) that frames the broader implications for many-body equilibrium and thermodynamic relativity.

QUENCHED REPRESENTATION AND INFLUENCE FUNCTIONAL

We begin by recapitulating the quenched representation introduced in Ref. [40]. To proceed directly from Sec. , we explicit the composite model. We denote the bare system Hamiltonian by H_Q and write

$$H_{\text{tot}} = H_Q + H_X + H_{\text{int}}, \quad (3)$$

where H_X is the bath Hamiltonian. We assume a factorizable interaction

$$H_{\text{int}} = f \otimes B, \quad (4)$$

where f acts on the system and B is a bath operator. We can define the reduced equilibrium operator (up to normalisation) by $\bar{\rho}_Q(\beta) \equiv \text{Tr}_X e^{-\beta H_{\text{tot}}}$. As shown in Ref. [40], this can be represented as a *quenched density*. This is an average over a stochastic propagator, given by:

$$\bar{\rho}_Q(\beta) = \mathbb{E}_\xi[U_\xi(\beta)], \quad (5)$$

$$U_\xi(\beta) \equiv \mathcal{T}_\tau \exp \left[- \int_0^\beta d\tau (H_Q + \xi(\tau)f) \right], \quad (6)$$

where $\xi(\tau)$ is a stochastic process whose statistics encode the bath correlations, and $\mathcal{T}_\tau$ denotes time-ordering in

imaginary time. Regardless of the precise form of bath and coupling, the reduced density must always be describable in this form.

This result can be understood intuitively by first observing that since the environment influence can only enter through the system coupling f , its influence can be captured by attaching a τ -dependent driving field to f [49, 50]. If this were a single deterministic field however, it would correspond to the bath exerting the *same* back-action history for every microscopic bath configuration. But Tr_X averages over many bath microstates in the thermal ensemble, and hence over many back-action histories. In this sense the partial trace is necessarily an average over histories in imaginary time, and the quenched representation simply makes this averaging explicit. For each realisation $\xi(\tau)$ the system evolves under an imaginary-time Hamiltonian $H_Q + \xi(\tau)f$; the bath is then recovered by averaging over $\xi(\tau)$ with a law chosen to reproduce the bath-induced correlations. In this sense $\xi(\tau)$ is not a physical external control field but an efficient parametrisation of the bath history $B(\tau)$ as seen through the coupling channel.

We may understand what formal properties are demanded of ξ by considering the influence functional it is required to match. Working in imaginary time, introduce the bath interaction picture:

$$B(\tau) \equiv e^{\tau H_X} B e^{-\tau H_X}, \quad \tau \in [0, \beta], \quad (7)$$

and define the bath thermal state $\rho_X \equiv e^{-\beta H_X} / Z_X$. For an arbitrary c -number source $j(\tau)$ coupled linearly to $B(\tau)$, the bath generates a (time-ordered) functional

$$\mathcal{Z}_X[j] \equiv \text{Tr}_X \left[\mathcal{T}_\tau \exp \left(- \int_0^\beta d\tau j(\tau) B(\tau) \right) \rho_X \right]. \quad (8)$$

Here $j(\tau)$ is introduced as an external c -number source used to generate ordered bath correlators by functional differentiation. More generally, $j(\tau)$ may be any object commuting with the bath algebra (e.g. a system operator tensored with $\mathbb{I}_X$). In the influence-functional derivation it is ultimately supplied by the system history, which becomes a c -number function in the path-integral representation. In the influence-functional approach, tracing out the bath produces precisely such a functional, evaluated on the system history through the coupling channel.

From this, the bath contribution to the effective Euclidean action can be written as [17, 18, 41]

$$\mathcal{F}[f] \equiv \mathcal{Z}_X[f], \quad \Phi[f] \equiv \log \mathcal{F}[f]. \quad (9)$$

A key structural fact is that $\Phi[f] = \log \mathcal{F}[f]$ is the *cumulant generating functional* of the bath operator $B(\tau)$ with respect to the thermal state. Concretely, the generalised (time-ordered) cumulant theorem implies the connected

expansion [51, 52]

$$\begin{aligned} \Phi[f] &= \sum_{n \geq 1} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \\ &\quad \times K^{(n)}(\tau_1, \dots, \tau_n) f(\tau_1) \cdots f(\tau_n), \end{aligned} \quad (10)$$

where the kernels $K^{(n)}$ are the *connected* (cumulant) bath correlators, defined via [53]

$$K^{(n)}(\tau_1, \dots, \tau_n) \equiv \langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \rangle_c. \quad (11)$$

The explicit connection back to Eq. (8) is given via its functional differentiation:

$$\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \rangle_c = (-1)^n \frac{\delta^n \log \mathcal{Z}_X[j]}{\delta j(\tau_1) \cdots \delta j(\tau_n)} \Big|_{j=0}. \quad (12)$$

The bath influence is then completely characterised by the hierarchy $\{K^{(n)}\}_{n \geq 1}$.

To connect the influence functional back to the quenched density, we use the fact that the bath influence depends on the system history only through the linear functional $\int_0^\beta d\tau f(\tau) B(\tau)$. One may therefore represent $\mathcal{F}[f]$ as the (generalised) characteristic functional of an auxiliary field $\xi(\tau)$ [42–44]:

$$\mathcal{F}[f] = \mathbb{E}_\xi \left[\exp \left(- \int_0^\beta d\tau \xi(\tau) f(\tau) \right) \right], \quad (13)$$

where $\mathbb{E}_\xi[\cdot]$ denotes averaging with respect to a (possibly complex) measure on ξ -histories chosen such that Eq. (13) holds.

Since $\xi(\tau)$ is a commuting c -number field, its n -point moments are symmetric under permutations of the time arguments. The influence kernels $K^{(n)}$ fix the *cumulants* of ξ via

$$\langle \xi(\tau_1) \cdots \xi(\tau_n) \rangle_c = K^{(n)}(\tau_1, \dots, \tau_n). \quad (14)$$

Consequently, the ordinary correlation functions of the noise are obtained from $\{K^{(m)}\}$ by the standard moment-cumulant relations:

$$\langle \xi(\tau_1) \cdots \xi(\tau_n) \rangle = \sum_{\pi \in \mathcal{P}_n} \prod_{C \in \pi} K^{(|C|)}(\{\tau_i\}_{i \in C}), \quad (15)$$

where $\mathcal{P}_n$ denotes the set of all partitions of the index set $\{1, \dots, n\}$, and C are the disjoint blocks of a given partition $\pi \in \mathcal{P}_n$. After shifting the mean so that $K^{(1)}(\tau) = \langle \xi(\tau) \rangle = 0$, one has (for example)

$$\langle \xi(\tau_1) \xi(\tau_2) \rangle = K^{(2)}(\tau_1, \tau_2), \quad (16)$$

$$\langle \xi(\tau_1) \xi(\tau_2) \xi(\tau_3) \rangle = K^{(3)}(\tau_1, \tau_2, \tau_3), \quad (17)$$

$$\begin{aligned} &\langle \xi(\tau_1) \xi(\tau_2) \xi(\tau_3) \xi(\tau_4) \rangle \\ &= K^{(4)}(\tau_1, \tau_2, \tau_3, \tau_4) \\ &\quad + K^{(2)}(\tau_1, \tau_2) K^{(2)}(\tau_3, \tau_4) \\ &\quad + K^{(2)}(\tau_1, \tau_3) K^{(2)}(\tau_2, \tau_4) \\ &\quad + K^{(2)}(\tau_1, \tau_4) K^{(2)}(\tau_2, \tau_3). \end{aligned} \quad (18)$$

QUENCHED DENSITY AND THE HAMILTONIAN OF MEAN FORCE

The Hamiltonian of mean force is, by definition, the operator whose Gibbs form reproduces the system's reduced equilibrium state. Equivalently, it is the *operator logarithm* of the unnormalised reduced equilibrium operator

$$\bar{\rho}_Q(\beta) \equiv \text{Tr}_X e^{-\beta H_{\text{tot}}}, \quad H_{\text{MF}}(\beta) \equiv -\frac{1}{\beta} \log \bar{\rho}_Q(\beta), \quad (19)$$

defined up to an additive multiple of the identity (fixed only when normalising the state).

The quenched representation in Eq. (5) supplies an exact stochastic parametrisation of the unnormalised mean-force Gibbs operator $\bar{\rho}_Q(\beta)$ by making the bath trace an explicit average over imaginary-time back-action histories. Combining Eq. (5) with Eq. (19) yields

$$H_{\text{MF}}(\beta) = -\frac{1}{\beta} \log \mathbb{E}_\xi[U_\xi(\beta)]. \quad (20)$$

Thus, constructing the mean-force Hamiltonian reduces to evaluating a stochastic average and then compressing the result via an operator logarithm. The conditions for being able to perform this average exactly is the focus of the present work.

To make the handling of this problem more concrete, we shall specialise the environment to the Caldeira-Leggett model, where the bath is a collection of harmonic oscillators ($H_X = \sum_k \omega_k b_k^\dagger b_k$) and the coupling is linear in bath coordinates ($B = \sum_k c_k x_k$). None of the results that follow are essentially dependent on this choice, and a generalisation to anharmonic environments is (relatively) straightforward. In the interests of comprehensibility however, we restrict our scope to quadratic environments. In this setting, the auxiliary field $\xi(\tau)$ becomes a stationary zero-mean Gaussian process completely characterized by its covariance

$$\mathbb{E}_\xi[\xi(\tau)\xi(\tau')] = K(\tau - \tau'). \quad (21)$$

The kernel $K(\tau)$ is determined by the bath spectral density $J(\omega) = \frac{\pi}{2} \sum_k \frac{c_k^2}{m_k \omega_k} \delta(\omega - \omega_k)$ via the relation [54]

$$K(\tau) = \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \frac{\cosh(\omega(\beta/2 - |\tau|))}{\sinh(\beta\omega/2)}. \quad (22)$$

A key property of this kernel is its integrated strength. Integrating Eq. (22) yields

$$\begin{aligned} \int_0^\beta d\tau K(\tau) &= \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \int_0^\beta d\tau \frac{\cosh(\omega(\beta/2 - |\tau|))}{\sinh(\beta\omega/2)} \\ &= \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \frac{2}{\omega} \\ &= 2\lambda, \end{aligned} \quad (23)$$

where λ is the explicit reorganisation energy. Consequently, the total variance of the integrated noise field $\Xi = \int_0^\beta d\tau \xi(\tau)$ grows linearly with inverse temperature:

$$\mathbb{E}_\xi[\Xi^2] = \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') = 2\beta\lambda. \quad (24)$$

In the case that the system and its coupling commute, $[H_Q, f] = 0$ and f is τ -independent in imaginary time. In this instance time ordering drops out, and the average is given by [40]:

$$\bar{\rho}_Q(\beta) = \exp \left[-\beta \left(H_Q - \frac{\kappa_0(\beta)}{2} f^2 \right) \right], \quad (25)$$

which in turn yields

$$H_{\text{MF}}(\beta) = H_Q - \frac{\kappa_0(\beta)}{2} f^2 + \frac{1}{\beta} \log Z_X(\beta) \mathbb{I}. \quad (26)$$

Notably this correction to $H_{\text{MF}}(\beta)$ is entirely *classical* [40]. This is hardly surprising, but emphasises that quantum effects *must* stem from non-commutativity. A truly quantum theory of thermodynamics therefore only makes meaningfully distinct predictions when $[H_Q, f] \neq 0$. In this case however, the noise enters through a noncommuting operator inside $\mathcal{T}_\tau$, rendering the question of averaging highly non-trivial. In the next section, we attack this problem directly, deriving conditions under which $H_{\text{MF}}(\beta)$ possesses a closed form.

GENERATING-FUNCTION STRUCTURE OF THE INFLUENCE OPERATOR

Quenched propagator and Gaussian resummation

The starting point is the quenched propagator introduced in Sec. . In the imaginary-time interaction picture with respect to H_Q , the unnormalised reduced equilibrium operator takes the form

$$\begin{aligned} \bar{\rho}_Q(\beta) &= e^{-\beta H_Q} \langle W_\xi(\beta) \rangle_\xi, \\ W_\xi(\beta) &\equiv \mathcal{T}_\tau \exp \left[- \int_0^\beta d\tau \xi(\tau) \tilde{f}(\tau) \right], \end{aligned} \quad (27)$$

where $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$ and the noise covariance is $\langle \xi(\tau)\xi(\tau') \rangle_\xi = K(\tau - \tau')$.

Because the bath is Gaussian, only the second cumulant of ξ is non-vanishing. Time-ordering introduces a non-trivial operator structure, but the key identity—proved in Appendix A—is that the Wick resummation of all pairings exponentiates exactly:

$$\langle W_\xi(\beta) \rangle_\xi = \exp \left[\frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')] \right]. \quad (28)$$

The proof is a direct application of Wick's theorem to the Dyson expansion of W_ξ : at order $2p$ in the expansion, there are $(2p)!/(2^p p!)$ distinct Wick pairings, each weighted by the two-point kernel $K(\tau_i - \tau_j)$. The prefactors from the Dyson expansion precisely cancel the pairing multiplicity, leaving only the singly-connected pairs.

We can exploit the symmetry $K(\tau - \tau') = K(\tau' - \tau)$ to fold the square domain onto the triangular one (see Appendix A), such that the exponent simplifies to

$$\Delta(\beta) \equiv \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau'), \quad (29)$$

and the averaged reduced state takes the product form

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q} e^{\Delta(\beta)}. \quad (30)$$

The influence exponent $\Delta(\beta)$ is quadratic in $\tilde{f}$ and depends on the bath exclusively through the two-point kernel K . Here the absence of higher-order bath objects is not an approximation, but an exact consequence of the Gaussian nature of the environment.

The ordered Green function and frequency compression

At this stage, the operator Δ is still formally defined as an integral over the bath degrees of freedom. There is however a particularly satisfying interpretation of this operator in which the effect of the bath is readily understood. To make this explicit, we work in the eigenbasis of H_Q . Let $H_Q|i\rangle = E_i|i\rangle$, so that the matrix elements of the interaction-picture operator are

$$\tilde{f}_{ij}(\tau) = \langle i|\tilde{f}(\tau)|j\rangle = f_{ij} e^{\omega_{ij}\tau}, \quad \omega_{ij} \equiv E_i - E_j. \quad (31)$$

Substituting into Eq. (29) and expanding in the eigenbasis, the influence exponent takes the spectral form

$$\Delta(\beta) = \sum_{i,\ell} \Delta_{i\ell} |i\rangle\langle\ell|, \quad (32)$$

where evaluating the matrix elements of the operator product yields

$$\Delta_{i\ell} = \sum_j f_{ij} f_{j\ell} G^>(\omega_{ij}, \omega_{j\ell}), \quad (33)$$

where all bath and temperature dependence has been absorbed into the *ordered Green function*

$$G^>(\omega_1, \omega_2) \equiv \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') e^{\omega_1\tau + \omega_2\tau'}. \quad (34)$$

In standard finite-temperature perturbation theory, analogous time-ordered integrals are evaluated using discrete Matsubara frequencies. Here, the integration instead

projects the bath response directly onto the Laplace frequencies set by the system's own energy differences; the superscript $>$ records that $\tau > \tau'$ throughout the integration domain.

The double integral in Eq. (34) collapses to a one-dimensional Laplace transform upon changing variables to $u = \tau - \tau'$ and $v = \tau'$:

$$G^>(\omega_1, \omega_2) = \frac{e^{\beta\omega_1} \mathcal{K}(\omega_2) - \mathcal{K}(\omega_1)}{\omega_1 + \omega_2}, \quad (35)$$

where

$$\mathcal{K}(\omega) \equiv \int_0^\beta du K(u) e^{\omega u} \quad (36)$$

is the bare Laplace transform of the bath kernel. Eq. (35) is an exact closed form valid for all $\omega_1 + \omega_2 \neq 0$. Importantly, it implies that for any transition between levels in the reduced system, the entirety of the bath's influence is encoded by $\mathcal{K}$ evaluated at the transition frequency.

The resonant channel $\omega_1 + \omega_2 = 0$ requires separate treatment. Taking the limit carefully yields

$$R(\omega) \equiv G^>(\omega, -\omega) = \int_0^\beta (\beta - u) K(u) e^{\omega u} du, \quad (37)$$

where the factor $(\beta - u)$ is the available imaginary time after a bath correlation at separation u ; the resonant kernel thus accumulates bath memory over the full thermal interval, regulated by the finiteness of β .

The structure of Eq. (33) admits a pleasingly simple diagrammatic reading, illustrated in Fig. 2. To isolate the physically distinct effects of the bath, we partition the influence exponent into its secular (diagonal) and mixing (off-diagonal) components, $\Delta = \Delta_{\parallel} + \Delta_{\perp}$.

The secular sector $\Delta_{\parallel} \equiv \sum_i \Delta_{ii} |i\rangle\langle i|$ singles out the *closed loops* $i \rightarrow j \rightarrow i$, for which $\omega_{ij} + \omega_{ji} = 0$ identically. The resonance condition forces the bath propagator onto $R(\omega_{ij})$, the object that accumulates bath memory over the whole thermal interval. The diagonal elements are thus the exact, non-perturbative *self-energy* dressing of the i -th energy level by all levels j it is coupled to:

$$\Delta_{ii} = \sum_j |f_{ij}|^2 R(\omega_{ij}). \quad (38)$$

Conversely, the mixing sector $\Delta_{\perp} \equiv \sum_{i \neq \ell} \Delta_{i\ell} |i\rangle\langle\ell|$ describes *open chains* $i \rightarrow j \rightarrow \ell$, where energy is not conserved at the intermediate vertex. Using the closed form in Eq. (35), these components can be expressed directly in terms of the Laplace-transformed bath kernel $\mathcal{K}(\omega)$:

$$\Delta_{i\ell} = \frac{1}{\omega_{i\ell}} \sum_j f_{ij} f_{j\ell} [e^{\beta\omega_{ij}} \mathcal{K}(\omega_{j\ell}) - \mathcal{K}(\omega_{ij})] \quad (i \neq \ell), \quad (39)$$

which encodes the coherent transitions between distinct energy levels mediated by the asymmetric fluctuations of the bath.

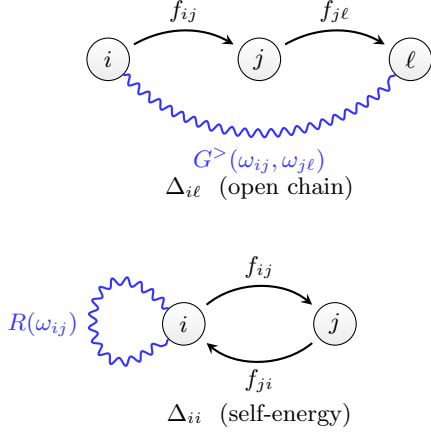


FIG. 2. Diagrammatic structure of the influence exponent $\Delta(\beta)$ in the H_Q eigenbasis. *Top*: The off-diagonal element $\Delta_{i\ell}$ (open chain) sums intermediate sites j , weighted by the bath propagator $G^>(\omega_{ij}, \omega_{j\ell})$ (wavy line) and two coupling insertions f (arrows). Energy is not conserved at the intermediate vertex. *Bottom*: The diagonal element Δ_{ii} (self-energy) is a closed loop $i \rightarrow j \rightarrow i$; the resonant condition $\omega_1 + \omega_2 = 0$ is satisfied, and the bath propagator collapses to $R(\omega_{ij})$.

KMS symmetry and the symmetrised reduced state

The bath kernel $K(\tau)$ satisfies the KMS periodicity condition $K(\tau) = K(\beta - \tau)$, which in Laplace space reads

$$\mathcal{K}(-\omega) = e^{-\beta\omega} \mathcal{K}(\omega). \quad (40)$$

Substituting into the closed form Eq. (35) and simplifying gives a corresponding symmetry of the ordered Green function:

$$G^>(-\omega_2, -\omega_1) = e^{-\beta(\omega_1 + \omega_2)} G^>(\omega_1, \omega_2). \quad (41)$$

Because transition frequencies telescope, $\omega_{ij} + \omega_{j\ell} = \omega_{i\ell}$, applying Eq. (41) to the matrix-element formula Eq. (33) immediately yields the *detailed-balance* constraint

$$\Delta_{i\ell} = e^{-\beta\omega_{i\ell}} \Delta_{i\ell}^*, \quad (42)$$

or in operator form $\Delta^\dagger = \Pi \Delta \Pi^{-1}$ where $\Pi \equiv e^{-\beta H_Q}$. Although Δ itself is not self-adjoint, this relation is a precise constraint: the product $\Pi \Delta$ is Hermitian.

This symmetry allows us to recast the unnormalised reduced state Eq. (30) into a manifestly positive, Hermitian form. Writing the bare Gibbs factor as $\Pi \equiv e^{-\beta H_Q}$ gives $\bar{\rho}_Q = \Pi e^\Delta$. By splitting $\Pi = \Pi^{1/2} \Pi^{1/2}$ and inserting the identity $\Pi^{-1/2} \Pi^{1/2} = \mathbf{1}$, we symmetrically distribute the bare weight:

$$\bar{\rho}_Q = \Pi^{1/2} (\Pi^{1/2} e^\Delta \Pi^{-1/2}) \Pi^{1/2}. \quad (43)$$

Applying the operator identity $A e^B A^{-1} = \exp(ABA^{-1})$ to the inner bracket yields

$$\bar{\rho}_Q = \Pi^{1/2} e^S \Pi^{1/2}, \quad S \equiv \Pi^{1/2} \Delta \Pi^{-1/2}, \quad (44)$$

where S is the *symmetrised influence operator*. Its matrix elements evaluate to $S_{i\ell} = e^{-\beta\omega_{i\ell}/2} \Delta_{i\ell}$, which by Eq. (42) satisfy $S_{i\ell}^* = S_{\ell i}$. Thus, the modified exponent S is strictly self-adjoint. The partition function of the reduced system follows by cyclicity of the trace:

$$Z_Q \equiv \text{Tr}[\bar{\rho}_Q] = \text{Tr}[\Pi e^S] = \sum_i e^{-\beta E_i} (e^S)_{ii}. \quad (45)$$

Since S is self-adjoint, $(e^S)_{ii} \geq e^{S_{ii}} = e^{\Delta_{ii}}$ by Jensen's inequality applied to the spectral weights $|\langle i|n\rangle|^2$, giving the bound

$$Z_Q \geq \sum_i e^{-\beta E_i + \Delta_{ii}}, \quad (46)$$

where the right-hand side is a Boltzmann sum with bath-shifted energies and equality holds when S is diagonal (i.e. when H_Q and Δ share an eigenbasis). The full result $Z_Q = \text{Tr}[\Pi e^S]$ captures all level-mixing corrections through the off-diagonal elements of S .

THE HAMILTONIAN OF MEAN FORCE

With this, we may now give the general expression for the Hamiltonian of Mean-Force.

$$-\beta H_{\text{MF}}(\beta) = \log(e^{-\beta H_Q} e^{\Delta(\beta)}). \quad (47)$$

It is worth pausing before the explicit construction to understand why this expression achieves exactness at all coupling strengths, where standard perturbative methods cannot. The conventional BCH and Magnus expansions are organised as power series in the system-bath coupling: each term involves the unbounded bath operator B , and the hierarchy of nested commutators generated by $H_{\text{int}} = \lambda f \otimes B$ has no natural truncation when λ is large. The polaron frame, HEOM hierarchies, and chain-mapping truncations are each attempts to reorder or re-express this divergent series in more tractable coordinates. They are in essence sophisticated fixes for the same underlying problem.

The quenched approach avoids the problem by design. The bath trace is performed *before* any expansion, via the privilege only a path integral provides. The price of this is that the reversion back to an operator picture requires stochasticity. For a quadratic environment, the Gaussian average $\langle W_\xi \rangle_\xi = e^\Delta$ is the cumulant theorem applied in its entirety, resumming every order of the noise at every coupling strength. What remains is $\Delta(\beta)$ — a system operator, entirely on the finite-dimensional Hilbert space of the system, built from the adjoint chain $\{f_n\} \subset \text{End}(\mathcal{H}_Q)$. The subsequent step, taking the BCH logarithm of $e^{-\beta H_Q} e^\Delta$, is therefore an algebraic operation on bounded operators within a finite-dimensional space. By the Cayley-Hamilton theorem, any polynomial in such operators terminates, and a closed form HMF becomes

automatic whenever the adjoint chain of f terminates. Exactness is not achieved by resumming a perturbative series; it follows from the observation that, for a finite-dimensional system, there is no need to begin one. If one permits poetic licence, the perturbation we expand is the Lie algebra of the model itself.

If H_Q and Δ commute, the logarithm would trivially give $-\beta H_Q + \Delta$. In general they do not, and the Baker–Campbell–Hausdorff (BCH) expansion

$$\log(e^A e^B) = A + \Phi(\text{ad}_A) B + O(B^2), \quad \Phi(x) \equiv \frac{x}{1 - e^{-x}}, \quad (48)$$

generates a tower of commutators of $A = -\beta H_Q$ acting on $B = \Delta$. To understand the structure of these corrections, we invoke the partition of the influence exponent into its secular and mixing parts, $\Delta = \Delta_{\parallel} + \Delta_{\perp}$. Because the secular piece $\Delta_{\parallel}$ strictly commutes with A , its adjoint action is annihilated ($\text{ad}_A(\Delta_{\parallel}) = 0$).

Consequently, the generating function evaluated at $x = 0$ yields $\Phi(0) = 1$, allowing the parallel component to push straight through the linear BCH expansion:

$$-\beta H_{\text{MF}} \approx (-\beta H_Q + \Delta_{\parallel}) + \Phi(\text{ad}_{-\beta H_Q}) \Delta_{\perp} + O(\Delta^2). \quad (49)$$

The secular zeroth-order shift $\Delta_{\parallel}$ thus groups directly with the bare system (reminiscent of the weak-coupling Lamb shift). Crucially however, while it acts geometrically like a classical energy shift, this term is still fundamentally *quantum* in origin because it is weighted by the bath Green functions, dynamically entangling the bare spectrum with the full spectral density of the environment. We can understand this directly by reference to Eq. (38). By defining a bath-dressed coupling operator $\tilde{f}$ with matrix elements $\tilde{f}_{ij} = f_{ij} \sqrt{R(\omega_{ij})}$, the secular exponent takes the exact quadratic form $\Delta_{\parallel} = (\tilde{f} \tilde{f}^{\dagger})_{\text{diag}}$. In the macroscopic limit where the system commutes with its coupling ($[H_Q, f] = 0$), the matrix f is entirely diagonal in the energy basis ($\omega_{ij} = 0$), and this dressed square collapses precisely to $\Delta_{\parallel} = \beta \lambda f^2$, where $R(0) = \beta \lambda$ is the explicit reorganisation energy defined in Eq. (23). The zeroth order BCH term therefore exactly recovers the classical commuting shift $H_{\text{MF}} = H_Q - \lambda f^2$ of Eq. (26), and at zeroth order the non-commuting correction can be obtained by replacing $f \rightarrow \tilde{f}$. The remaining coherence-generating effects are then driven exclusively by the action of Φ on the perpendicular, off-diagonal open-chain fragments $\Delta_{\perp}$.

There is a satisfyingly clean expression for these non-commuting corrections in the system energy eigenbasis. The off-diagonal basis operators $|i\rangle\langle j|$ are exact eigenoperators of the adjoint map: $\text{ad}_{-\beta H_Q} |i\rangle\langle j| = -\beta [H_Q, |i\rangle\langle j|] = -\beta \omega_{ij} |i\rangle\langle j|$. Consequently, the entire infinite tower of nested commutators generated by $\Phi(\text{ad}_{-\beta H_Q})$ trivially evaluates to a simple scalar multiplication for each matrix element of the perpendicular

fragment:

$$\Phi(\text{ad}_{-\beta H_Q}) \Delta_{\perp} = \sum_{i \neq j} \Phi(-\beta \omega_{ij}) \Delta_{ij} |i\rangle\langle j|, \quad (50)$$

where $\Phi(x) = x/(1 - e^{-x})$. The off-diagonal elements of the linear-in- Δ effective Hamiltonian are therefore simply the bare bath-induced transitions Δ_{ij} , parametrically reweighted by the thermal asymmetry factor $\Phi(-\beta \omega_{ij}) = -\beta \omega_{ij}/(1 - e^{\beta \omega_{ij}})$.

This scalar evaluation exhibits a manifestly physical symmetry. If we reconstruct the effective matrix element using the symmetrised influence operator $S_{ij} = e^{-\beta \omega_{ij}/2} \Delta_{ij}$, the thermal asymmetry factor exactly absorbs the detailed-balance twist. The (off-diagonal) components on the effective Hamiltonian are therefore simply the bare bath-induced transitions Δ_{ij} , parametrically reweighted by the thermal asymmetry factor $\Phi(-\beta \omega_{ij}) = -\beta \omega_{ij}/(1 - e^{\beta \omega_{ij}})$:

$$\begin{aligned} (H_{\text{MF}})_{ij} &= -\frac{1}{\beta} \Phi(-\beta \omega_{ij}) e^{\beta \omega_{ij}/2} S_{ij} \\ &= -\frac{\omega_{ij}/2}{\sinh(\beta \omega_{ij}/2)} e^{-\beta \omega_{ij}/2} \Delta_{ij}. \end{aligned} \quad (51)$$

Because $x/\sinh(x)$ is an explicitly even geometric function of the energy gap ω_{ij} , Eq. (51) guarantees that the linearly truncated effective Hamiltonian is perfectly Hermitian. Moreover, this exact thermal weighting reveals that bath-induced coherent mixing between levels is heavily exponentially suppressed for high-frequency transitions ($|\beta \omega_{ij}| \gg 1$), dynamically localising the leading mean force corrections to the thermodynamic energy shell.

This fortuitous cancellation reveals a deep structural connection between quantum thermodynamics and Lie algebra. The fundamental detailed balance condition of the bath imposes a thermal asymmetry $e^{-\beta \omega}$ on the bare transition amplitudes. Conversely, the BCH expansion - a purely abstract property of non-commuting logarithms - generates the parametric weighting $\Phi(-\beta \omega) \propto (1 - e^{\beta \omega})^{-1}$. The fact that the Lie-algebraic geometry of the thermal exponential perfectly mirrors - and thereby absorbs - the thermodynamic detailed-balance twist of the environment is striking, but on reflection perhaps unsurprising. After all, the conditions of detailed balance is imposed from thermality, and any condition relating the additive (energetic) and multiplicative (entropic) compositions of structures must ultimately run through the logarithm.

SPIN-BOSON MODEL

The spin-boson model [48, 54] - a two-level system coupled linearly to a bosonic bath - occupies a unique position in the theory of open quantum systems. It

is the simplest model that captures the full competition between quantum coherence and environmental dissipation, and it arises as a controlled description across condensed matter, quantum optics, and quantum chemistry, from Josephson junctions and impurity problems to photosynthetic energy transfer and qubit decoherence. Despite more than four decades of investigation using path integrals, renormalisation group methods, numerical renormalisation group, and tensor-network approaches, a general closed-form solution for the reduced equilibrium state at arbitrary coupling and temperature remains unknown [48, 54–56]. Exact results are confined to special limits: weak coupling (Redfield/Born–Markov), the scaling limit near the Toulouse point, and the zero-temperature ground state in certain parameter regimes. The exact thermodynamics of the intermediate-coupling, finite-temperature regime remains inaccessible to standard perturbative techniques.

In some sense this is an embarrassment for the field. Regardless of whether one marks the model’s inception in the 1960s with the Kondo model [57], or its formal coining in the 1980s [48], it is a problem that has remained open for decades. It is high time for the case to be closed. To do so, we now derive the exact solution for the reduced equilibrium state of the spin-boson model, with the caveat that we deal only with finite temperature. While the necessary machinery for the $\beta \rightarrow \infty$ ground state solution is implicit in the present work, we defer its full treatment to future work.

Let us first set our conventions. As in Sec. , we work in the qubit energy basis ($\hat{\mathbf{n}}_s = \hat{\mathbf{z}}$) during derivation, then state the final geometry in the coupling plane spanned by $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}$. We take

$$H_Q = \frac{\omega_q}{2} \hat{\mathbf{n}}_s \cdot \boldsymbol{\sigma}, \quad f = \hat{\mathbf{f}} \cdot \boldsymbol{\sigma}, \quad (52)$$

with $\hat{\mathbf{f}} = (s \cos \phi, s \sin \phi, c)$, $c \equiv \cos \theta$, $s \equiv \sin \theta$. The dimensionless coupling amplitude g is suppressed in intermediate expressions and reinserted where scaling is discussed.

Bath kernels and the symmetrised influence state

The interaction-picture coupling operator is

$$\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q} = c \sigma_z + f_- e^{\omega_q \tau} \sigma_+ + f_+ e^{-\omega_q \tau} \sigma_-, \quad (53)$$

with $f_{\pm} = s e^{\pm i \phi}$. The required $SU(2)$ algebra is standard and relegated to App. C. The final result is that all bath dependence enters through two real response channels

$$\begin{aligned} \Sigma_z &\equiv \frac{1}{2} [R(\omega_q) - R(-\omega_q)], \\ \Sigma_{\perp} &\equiv \frac{4}{\omega_q} \left[\cosh\left(\frac{\beta \omega_q}{2}\right) \mathcal{K}(0) - e^{-\beta \omega_q/2} \mathcal{K}(\omega_q) \right]. \end{aligned} \quad (54)$$

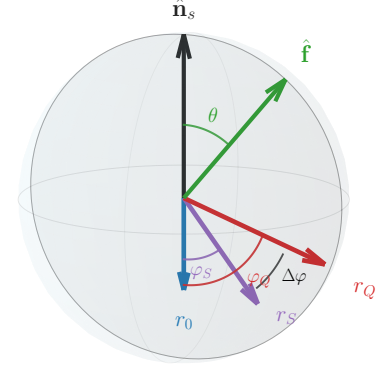


FIG. 3. Bloch-sphere geometry with the three radii used in the text. The bare thermal state has radius r_0 (blue), the symmetrised influence state $\rho_S \propto e^S$ has radius r_S and signed tilt φ_S (purple), and the final reduced state has radius r_Q and tilt φ_Q (red). The black arc marks the relative rotation $\Delta\varphi = \varphi_Q - \varphi_S$.

Using the symmetric decomposition $\bar{\rho}_Q = \Pi^{1/2} e^S \Pi^{1/2}$, $\Pi = e^{-\beta H_Q}$, the symmetrised influence operator is

$$S = \Delta_0 \mathbb{I} + s^2 \Sigma_z \sigma_z - s c \Sigma_{\perp} (\cos \phi \sigma_x + \sin \phi \sigma_y). \quad (55)$$

The influence operator can then be written as $M \equiv S - \Delta_0 \mathbb{I} = \mathbf{m}_S \cdot \boldsymbol{\sigma}$. After some squinting, one sees that its may be written in terms of the perpendicular to the coupling vector, i.e. $\hat{\mathbf{f}}_{\perp} = (-c \cos \phi, -c \sin \phi, s)$. So the influence Bloch vector is

$$\mathbf{m}_S = s \Sigma_{\perp} \hat{\mathbf{f}}_{\perp} + s^2 (\Sigma_z - \Sigma_{\perp}) \hat{\mathbf{n}}_s, \quad (56)$$

The *influence magnitude* is then

$$\chi = \|\mathbf{m}_S\| = |s| \sqrt{c^2 \Sigma_{\perp}^2 + s^2 \Sigma_z^2} \quad (57)$$

note that the magnitude depends on the *absolute value* of $s = \sin \theta$. Unless otherwise stated, all results shall assume $s > 0$.

The *influence state* is now naturally given by $\rho_S = e^S / \text{Tr}(e^S)$. Evaluating this yields

$$\rho_S = \frac{1}{2} [\mathbb{I} + \tanh \chi \hat{\mathbf{m}}_S \cdot \boldsymbol{\sigma}], \quad \hat{\mathbf{m}}_S = \mathbf{m}_S / \chi. \quad (58)$$

In this form, the geometry of the influence state on the Bloch sphere is clear. It has radius r_S

$$r_S = \tanh \chi \quad (59)$$

while its angular content in the plane is determined by the bath. The component perpendicular to the coupling vector is set by the transverse (off-resonant) response, while the parallel component is set by the anisotropy between the longitudinal (resonant) and transverse responses. In the isotropic kernel limit $\Sigma_\perp = \Sigma_z$, that parallel piece vanishes and e^S points exactly perpendicular to $\hat{\mathbf{f}}$. The bath anisotropy is directly encoded in the *signed tilt* of ρ_S from the bare axis ($-\hat{\mathbf{n}}_s$) via the relation

$$\tan \varphi_S = -\frac{c}{s} \frac{\Sigma_\perp}{\Sigma_z}. \quad (60)$$

The sign of $\chi \propto s$ carries orientability: $(\chi, \hat{\mathbf{m}}_S)$ and $(-\chi, -\hat{\mathbf{m}}_S)$ are equivalent charts of the same physical ρ_S . This discrete orientability is benign at the ρ_S stage but becomes consequential after recombination with the bare Gibbs factor.

Reduced state in signed Bloch form

With ρ_S in hand, the reduced state follows by recombination with the bare Gibbs factor, $\rho_Q = \Pi^{1/2} \rho_S \Pi^{1/2}$. It takes the same Bloch form,

$$\rho_Q = \frac{1}{2} [\mathbb{I} + r_Q \hat{\mathbf{m}}_Q \cdot \boldsymbol{\sigma}], \quad (61)$$

where $r_Q = \sqrt{m_z^2 + m_\perp^2}$ and the components are resolved along and perpendicular to the bare axis via the tilt angle φ_Q :

$$m_z = r_Q \cos \varphi_Q, \quad m_\perp = r_Q \sin \varphi_Q. \quad (62)$$

We can define the effective longitudinal scale by the projection of the influence state onto the bare axis. Defining the factor

$$\gamma(\chi) \equiv \tanh \chi / \chi \quad (63)$$

the effective longitudinal scale is given by

$$\Theta \equiv \operatorname{arctanh}(\gamma(\chi) s^2 \Sigma_z) - \frac{\beta \omega_q}{2}. \quad (64)$$

The explicit solution is then

$$m_z = \tanh \Theta, \quad (65)$$

$$m_\perp = \tan \varphi_S \left(m_z \cosh \frac{\beta \omega_q}{2} + \sinh \frac{\beta \omega_q}{2} \right), \quad (66)$$

where φ_S is the tilt angle of the influence state, Eq. (60). The structure is clear: m_z is a thermal magnetisation evaluated at an *effective* longitudinal scale Θ , which encodes the bath-renormalised temperature, while $m_\perp$ is locked to m_z through the bath-induced tilt φ_S . Defining the bare system radius as $r_0 = \tanh(\beta \omega_q / 2)$, the final tilt angle is

$$\tan \varphi_Q = \tan \varphi_S \frac{m_z + r_0}{\sqrt{1 - r_0^2} m_z}, \quad (67)$$

and the radius satisfies

$$1 - r_Q^2 = \frac{(1 - r_0^2)(1 - r_S^2)}{(1 + r_0 r_S \cos \varphi_S)^2}. \quad (68)$$

Thus recombination acts as a *two-step deformation* of ρ_S : a shift in radius governed by the competition between r_0 and r_S , controlled ultimately by the influence magnitude and a rotation of the tilt angle from (definitionally) to φ_Q . These effects are ultimately controlled by just three factors - the angle of the coupling, the anisotropy of the bath response, and its effective magnitude χ .

THERMAL RENORMALISABILITY, DUALITY, AND RELATIVITY

The influence state ρ_S enjoys a discrete orientability symmetry: as noted above, the pair $(\chi, \hat{\mathbf{m}}_S)$ and $(-\chi, -\hat{\mathbf{m}}_S)$ are equivalent charts of the same physical state. At the level of ρ_S this is entirely benign — both charts represent the same point on the Bloch sphere. After recombination with the bare Gibbs weight, however, this symmetry is *broken*. The bare factor $\Pi^{1/2}$ carries a definite orientation along $-\hat{\mathbf{n}}_s$, and the two orientability charts of ρ_S couple to it differently. The result is a genuine *representational duality* in the reduced state: two inequivalent thermal descriptions of the same physics.

To expose this duality it is useful to recast the result as a thermal renormalisation. Working in the bare-system orientation, we observe that the longitudinal component $m_z = \tanh \Theta$ has exactly the form of a thermal magnetisation. This motivates defining a *renormalised inverse temperature*

$$\bar{\beta} \equiv -\frac{2\Theta}{\omega_q}, \quad (69)$$

so that $m_z = \tanh(-\bar{\beta} \omega_q / 2)$, with corresponding renormalised radius $\bar{r} \equiv |m_z|$. The bath's longitudinal influence then acts as a pure shift of the inverse temperature from the bare β to

$$\bar{\beta} = \beta - \frac{2}{\omega_q} \operatorname{arctanh}(\gamma(\chi) s^2 \Sigma_z). \quad (70)$$

The transverse bath response captures the remainder of the influence by tilting the renormalised state. This renormalised tilt angle is definitionally equivalent to the reduced state's tilt: $\bar{\varphi} = \varphi_Q$. This raises an immediate question: why bother with the relabelling? The answer is that the physical content depends critically on the *sign* of Θ , and this is precisely where the broken orientability of the influence state reasserts itself.

The two physically transparent special cases illustrate the structure cleanly. For *pure longitudinal coupling* ($s = 0$), the influence state is maximally mixed ($\chi = 0$, $\rho_S = \mathbb{I}/2$), recombination reproduces the bare Gibbs state, and

both dual charts collapse to $\bar{\beta} = \beta$: this is the self-dual fixed point with no thermal renormalisation. For *pure transverse coupling* ($\theta = \pi/2$, $c = 0$), the longitudinal response vanishes identically ($\Sigma_z \mapsto 0$), the tilt angle $\varphi_Q \rightarrow 0$, and the renormalised temperature reduces to

$$\bar{\beta}|_{\Sigma_z=0} = \beta \pm \frac{2\Sigma_\perp}{\omega_q}, \quad (71)$$

where the two signs correspond to the two dual charts. A purely transverse coupling is therefore nothing more than a shift in effective temperature, with the direction of the shift set by the relative orientation of coupling and bare-system axis.

The key observation is that the operation $\theta \mapsto \pi - \theta$ (i.e. reflecting the coupling vector about the equatorial plane, $c \rightarrow -c$, $s \rightarrow s$) admits two complementary readings. From the geometric side, it is a *global phase*: since $\chi \propto |s|$, the influence radius $r_S = \tanh \chi$ is invariant, and the operation is simply a relabelling of the coupling orientation that leaves ρ_S unchanged. But to maintain exactly the same influence state ρ_S when passing from θ to $\pi - \theta$, the bath must compensate: its longitudinal susceptibility must undergo $\Sigma_z \mapsto -\Sigma_z$. In this second reading, the same discrete transformation is a genuine *symmetry-breaking operation* — a change in the bath susceptibilities, not a geometric relabelling. The duality between the two reduced-state representatives is the footprint of this ambiguity: the influence state ρ_S cannot distinguish the two, but after recombination with the bare Gibbs weight, ρ_Q can. Under $\Sigma_z \mapsto -\Sigma_z$ at fixed θ , the effect on the longitudinal scale is $\text{arctanh}(\gamma s^2 \Sigma_z) \mapsto -\text{arctanh}(\gamma s^2 \Sigma_z)$, giving

$$\Theta \mapsto -\Theta - \beta\omega_q, \quad \bar{\beta} \mapsto 2\beta - \bar{\beta}, \quad \varphi_Q \mapsto \pi - \varphi_Q. \quad (72)$$

The two reduced states are genuinely distinct - a reflected tilt and a dual temperature - but are possess a natural *duality* structure. At the level of $\bar{\beta}$, we have the dual charts:

$$\bar{\beta}^\vee = 2\beta - \bar{\beta}, \quad \bar{\beta} + \bar{\beta}^\vee = 2\beta, \quad (73)$$

with $\bar{\beta} = \beta$ (i.e. $\Theta = 0$) as the self-dual symmetry point. One orientability chart renormalises the temperature upward; the other renormalises it downward. The dual mappings for the renormalised radius and tilt angle take the equally compact forms

$$\text{arctanh } \bar{r} + \text{arctanh } \bar{r}^\vee = \beta\omega_q, \quad \varphi_Q + \varphi_Q^\vee = \pi, \quad (74)$$

mirroring the structure of Eq. (73): the dual pairs each sum to their natural full-scale value (2β , $\beta\omega_q$, and π respectively). Since $\text{arctanh } r = \frac{1}{2} \ln \frac{1+r}{1-r}$, the radius relation is equivalently a constraint on the product of polarisation ratios,

$$\frac{1+\bar{r}}{1-\bar{r}} \cdot \frac{1+\bar{r}^\vee}{1-\bar{r}^\vee} = \left(\frac{1+r_0}{1-r_0} \right)^2, \quad (75)$$

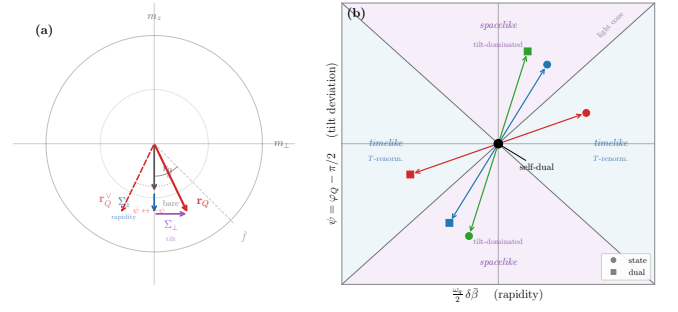


FIG. 4. Thermodynamic Minkowski diagram in the centred thermal coordinates $(\omega_q \delta\bar{\beta}/2, \psi)$ (Eq. (76)). The dual map acts as a point reflection about the self-dual origin ($\bullet$), exchanging each state (circles) with its dual (squares). Temperature renormalisation dominates in the timelike (blue) sectors; tilt dominates in the spacelike (purple) sectors.

making the opposite-scaling property explicit: if $\bar{r} > r_0$ (temperature renormalised downward), then necessarily $\bar{r}^\vee < r_0$ (renormalised upward), and vice versa. The structure is manifestly relativistic: identifying $\text{arctanh } \bar{r}$ with a rapidity, the dual constraint is a fixed sum of rapidities — precisely the kinematics of a Lorentz boost, with the bare Gibbs factor playing the role of the boost. The iso-dual curves are rectangular hyperbolas (cf. the dual-pair lines in Fig. 4). All three dual maps are *involution* — applying either map twice returns the original value — with the common self-dual fixed point at $\bar{\beta} = \beta$, $\bar{r} = r_0$, $\varphi_Q = \pi/2$.

The tilt angle offers a complementary picture. Defining the centred deviation $\psi \equiv \varphi_Q - \pi/2$, the dual map simply sends $\psi \mapsto -\psi$: the two representations sit at equal and opposite angles about the self-dual point. This motivates a final interpretation of the duality structure created by the bath symmetry breaking. In the relativistic language, the rapidity (radius) and tilt (angle) play conjugate roles — the rapidities sum to a positive constant (timelike), while the centred tilts are equal and opposite (spacelike). Together they equip the dual structure with the signature of a Minkowski metric, with the self-dual fixed point at its light-cone vertex. Using $\eta \equiv \text{arctanh } \bar{r} = \bar{\beta}\omega_q/2$ and centred coordinates $\delta\bar{\beta} \equiv \bar{\beta} - \beta$, $\psi \equiv \varphi_Q - \pi/2$, the flat metric and its preserved interval take the explicit form

$$ds^2 = \left(\frac{\omega_q}{2} \right)^2 d\bar{\beta}^2 - d\varphi_Q^2, \quad s^2 = \left(\frac{\omega_q \delta\bar{\beta}}{2} \right)^2 - \psi^2, \quad (76)$$

with the dual map acting as the point reflection $(\delta\bar{\beta}, \psi) \mapsto (-\delta\bar{\beta}, -\psi)$ about the origin. States with $s^2 > 0$ are “timelike”: temperature renormalisation dominates; states with $s^2 < 0$ are “spacelike”: the tilt dominates. The light cone $|\omega_q \delta\bar{\beta}/2| = |\psi|$ divides these regimes and passes through the self-dual fixed point.

It feels somewhat glib to observe a rectangular hyperbola and immediately declare a relativistic structure, but

in the present case it is difficult to avoid. The thermodynamic description of black holes has been long established, and the present argument simply repays the favour from the opposite orientation. In this language the role of *mass* is played by

the bare transition frequency ω_q : it is the single scale of the two-level spectrum, and it sets the “speed” of the timelike coordinate in Eq. (76). Different values of ω_q correspond to different families of hyperbolas — different mass shells.

The generalisation to an N -level system is immediate in spirit, though with an important refinement: there is still only *one* physical temperature β . The $K = N(N-1)/2$ transition frequencies $\{\omega_k\}$ each carry their own renormalised inverse temperature $\bar{\beta}_k(\beta)$, but these are not independent — they all descend from β through the bath’s sector-specific renormalisations. Defining the *thermal renormalisation tensor*

$$\mathcal{T}_k \equiv \frac{\partial \bar{\beta}_k}{\partial \beta}, \quad (77)$$

the physical metric lives on the image of the map $\beta \mapsto \{\bar{\beta}_k(\beta)\}$. The timelike sector collapses back to a single coordinate β , dressed by $\mathcal{T}$:

$$ds^2 = \left[\sum_k \left(\frac{\omega_k}{2} \right)^2 \mathcal{T}_k^2 \right] d\beta^2 - \sum_k d\psi_k^2. \quad (78)$$

The bracketed coefficient — a weighted sum of squared renormalisation Jacobians — is the N -level generalisation of $(\omega_q/2)^2$ and constitutes the stress-energy content: the spectrum $\{\omega_k\}$ sets the mass scale of each sector, while $\mathcal{T}_k$ encodes how strongly the bath couples each transition to the physical temperature. The bath-induced angles $\psi_k = \varphi_{Q,k} - \pi/2$ supply the spacelike (momentum) content. A purely longitudinal coupling concentrates the “matter” on the timelike axis; general couplings distribute it across the light cone. The transition to a general-relativistic language is the passage to systems where the $\mathcal{T}_k$ vary across parameter space — precisely many-body or multi-mode environments where the coupling geometry itself becomes state-dependent. Amusingly, this apparent linearisation of the relativistic field theory is a natural consequence of the one *nonlinear* operation in quantum theory. The trace. A formal proof of this mapping is left as an exercise to the reader. We expect this to lie within the capacities of even the least creatively burdened physicist.

PHYSICAL REGIMES OF THE MEAN-FORCE STATE

Even in this simple case, the physics encoded in the mean force state by its dependence on temperature and bath coupling is startlingly rich. To expose this, we first

commit to a specific bath spectral density for numerics. We choose the Ohmic spectral density

$$J(\omega) = \alpha \omega e^{-\omega/\omega_c}, \quad \omega \in [0, 2\omega_q], \quad (79)$$

with $\alpha = 1$ and $\omega_c = 5\omega_q$. Unless otherwise stated, we further specialise to $\theta = \pi/4$ throughout. We emphasise here that the choice of spectral shape is not fundamental, as it only controls the way in which χ runs with temperature.

This apparently innocuous expression encodes the entire non-perturbative recombination of bare Hamiltonian and influence, and functions as an *order* parameter for the crossover between weak and strong coupling regimes. The reduced equilibrium state transitions between these regimes as χ passes through unity,

$$\chi(\beta, g, \theta) \sim 1 \quad \Longleftrightarrow \quad g^2 \sim \chi_0(\beta, \theta)^{-1}. \quad (80)$$

This crossover condition $\chi(\beta, g, \theta) = 1$ therefore defines a coupling-dependent temperature scale

$$g_\star(\beta) = \chi_0(\beta, \theta)^{-1/2}, \quad (81)$$

and an equivalent crossover temperature $\beta_\star(g)$. Notably, these dual crossover values are an exact property of the solution.

The function $\gamma(\chi)$ is elementary but worth dwelling on. It is monotone decreasing from $\gamma(0) = 1$ to $\gamma(\infty) = 0$ with its inflection near $\chi \approx 1.2$, and admits clean asymptotic forms:

$$\chi \ll 1 : \quad \gamma \simeq 1 - \frac{\chi^2}{3} \quad (\text{weak coupling}), \quad (82)$$

$$\chi \gg 1 : \quad \gamma \simeq \chi^{-1} = (g^2 \chi_0)^{-1} \quad (\text{ultrastrong}). \quad (83)$$

In the weak-coupling regime the state tracks the bare Gibbs thermal state with $O(g^2)$ corrections, as expected. Notably, as shown in Fig. 5, the crossover between regimes depends on *both* temperature and coupling strength. This means that two distinct regimes emerge in the limit of scaling g and β , wholly dependent on which side of the crossover is taken.

Notably, these critical temperatures and couplings have an *involution* relationship, in the sense that

$$\beta_\star(g_\star(\beta)) = \beta, \quad g_\star(\beta_\star(g)) = g. \quad (84)$$

This relationship between temperature and coupling means the crossover can be interpreted as the *self-dual* line in the (β, g) plane, as explored in Fig. 6. This line represents a coupling-dependent temperature scale $\beta_\star(g)$ (or equivalently $g_\star(\beta)$), where different parameter scaling relationships apply.

State purity

The purity of the reduced qubit state is $\mathcal{P}_Q = \frac{1}{2}(1 + r_Q^2)$, with $r_Q^2 = |\mathbf{r}_Q|^2$. From this we can immediately establish a universal result in the direct representative:

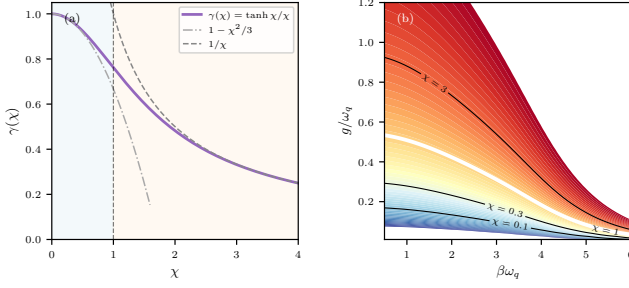


FIG. 5. Universal crossover for the qubit mean-force state. (a) Crossover function $\gamma(\chi) = \tanh \chi / \chi$ interpolating between the weak-coupling approximant $1 - \chi^2/3$ and the ultra-strong form $1/\chi$. Shaded regions mark the weak (blue) and strong (orange) regimes separated by $\chi = 1$. (b) Dimensionless coupling $\chi(\beta, g) = g^2 \chi_0(\beta)$ as a contour map. Vertical trajectories (fixed g) will crossover at its corresponding $\beta_*(g)$.

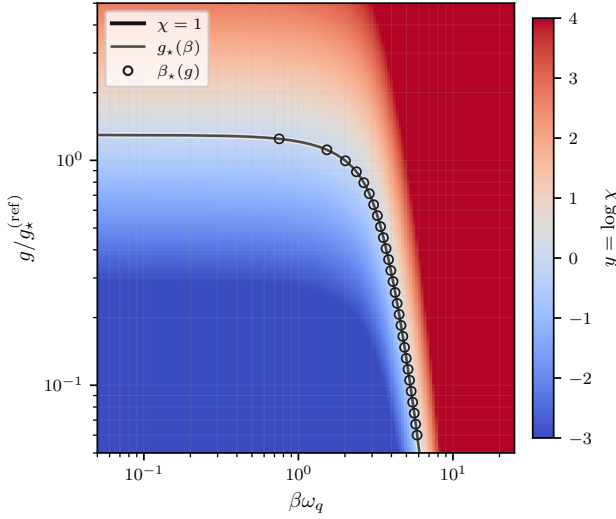


FIG. 6. Involutive crossover geometry generated by $\chi(\beta, g) = g^2 \chi_0(\beta)$. The heatmap shows $y = \log \chi$ in the (β, g) plane. The solid black line marks the crossover locus $\chi = 1$, which can be constructed either as a coupling-dependent temperature $\beta_*(g)$ (circles) or a temperature-dependent coupling $g_*(\beta)$ (solid grey line). This involutive relationship defines a partition between the weak-coupling (blue) and strong-coupling (red) scaling regimes.

Direct-representative bath purification.

For any positive spectral density $J(\omega) \geq 0$ and any $\beta > 0$: (i) $r_Q^2(g, \beta)$ is strictly increasing in g for $g \geq 0$, and (ii) for all $g > 0$,

$$r_Q^2(g, \beta) > r_0^2(\beta) \equiv \tanh^2\left(\frac{\beta\omega_q}{2}\right), \quad (85)$$

with equality only at $g = 0$.

Proof sketch. Part (i) follows immediately from $r_Q = \tanh \chi = \tanh(g^2 \chi_0(\beta))$: since $\tanh$ is strictly increasing

and $g^2 \chi_0$ is strictly increasing in g , so is r_Q .

Part (ii) (excess over bare) uses the non-negativity of the resonant (longitudinal) dressing Σ_z . From the explicit integral form of the response kernels (cf. App. C), one has

$$\Sigma_z = \frac{1}{\pi} \int_0^\beta u(\beta - u) K(u) \frac{\sinh(\omega_q u)}{u} du. \quad (86)$$

Since $K(u) \geq 0$ for a physical spectral density, and the remaining factors are non-negative on the integration domain, we have $\Sigma_z > 0$ for all $\beta > 0$.

Substituting the exact components $m_z = \tanh \Theta$ and $m_\perp$ from Eq. (66), the purity excess $\Delta r_Q^2 = r_Q^2 - \tanh^2(\beta\omega_q/2)$ is strictly positive whenever $\gamma s^2 \Sigma_z > 0$. Physically, the longitudinal bath channel reinforces polarisation and raises purity above the bare value. $\square$

In the limit $\beta \rightarrow 0$ (infinite temperature), the bare state is maximally mixed, yet the mean-force state has

$$r_Q|_{\beta\omega_q \rightarrow 0} = \tanh \chi > 0, \quad (87)$$

This result is independent of the bandwidth or spectral shape of $J(\omega)$: widening the bath increases $\chi_0(\beta)$ (driving the crossover to weaker g) but cannot reverse the purity inequality. Purity inversion would require reversing the sign of the longitudinal response ($\Sigma_z < 0$), which is impossible for any physical (positive) spectral density.

Figure 7 confirms this numerically. Panel (a) shows $r_Q(g/g_*)$ for four temperatures. Every trajectory lies strictly above its own r_0 reference line for all $g > 0$, confirming Eq. (85). Panel (b) shows $\Delta r_Q^2 = r_Q^2 - \tanh^2(\beta\omega_q/2)$ over the full (g, β) plane. Enhancement peaks at large g and high temperature (upper-right corner), and falls to near zero at low temperature (left edge) where the bare state is already nearly pure. The black dashed contour $\chi = 1$ marks the crossover at which purification accelerates steeply.

This statement needs one caveat. It applies to the *direct* representative (β, g) . Under the strong-weak representative involution $g^\vee = g_*^2/g$, one has $\chi^\vee = 1/\chi$, so

$$r_Q^\vee(\beta, g) = \tanh(\chi^\vee) = \tanh(1/\chi). \quad (88)$$

Hence a highly pure direct strong-branch point maps to a weak-branch dual point with reduced $r_Q^\vee$. This is representative duality, not a contradiction of Eq. (85) at fixed physical parameters.

Quantum enhancement of heat engines

An important feature of the crossover regime is that it demarcates where the state is maximally sensitive to bath coupling. This is seen by considering $\partial_g \varphi_Q$ and $\partial_g r_Q$, which quantify how rapidly the Bloch vector tilts and purifies per unit change in g . As Fig. 8 shows, both peak

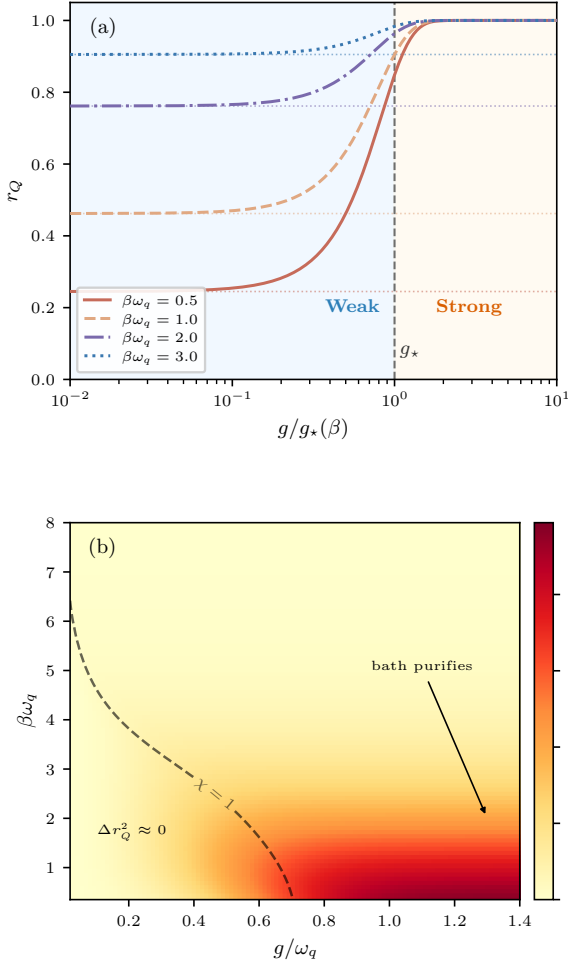


FIG. 7. Purity of the qubit mean-force state. **(a)** Reduced-state Bloch radius r_Q versus scaled coupling $g/g_*(\beta)$ for four temperatures. Dotted horizontals show the bare thermal purity $r_0 = \tanh(\beta\omega_q/2)$; coupling always increases purity (Eq. (85)), most visibly at high temperature where the bare state is near-maximally-mixed. Shaded regions mark the weak (blue) and strong (orange) coupling regimes. **(b)** Purity excess $\Delta r_Q^2 = r_Q^2 - \tanh^2(\beta\omega_q/2)$ in the (g, β) plane. The black dashed contour marks the crossover locus $\chi = 1$: purification is strongest at large g and high temperature and saturates as $r_Q \rightarrow 1$ deep in the strong-coupling regime.

near the crossover $\chi = 1$. $\partial_g \varphi_Q$ maximises at $g \approx 0.65 g_*$ (at $\chi_{\text{peak},\varphi} \approx 0.42$) and $\partial_g r_Q$ maximises at $g \approx 0.92 g_*$ ($\chi_{\text{peak},r} \approx 0.85$). The locations are approximately pinned at universal values of χ independent of temperature, as demonstrated by the alignment of peaks when plotted as a function of the rescaled coupling $g/g_*(\beta)$. The crossover region $\chi \sim 1$ is therefore where bath coupling yields the greatest rate of return: both the onset of coherence (tilt of the Bloch vector away from the bare thermal axis) and the onset of purification are most rapid here, before both effects saturate as $\chi \rightarrow \infty$. This suggests the crossover regime is the most operationally leveragable for

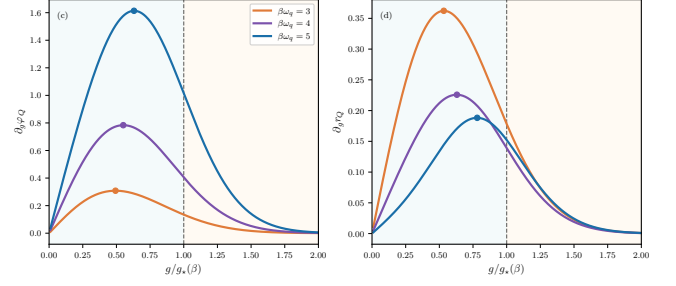


FIG. 8. Coherence and purity susceptibilities. **(c)** Reduced-state tilt susceptibility $\partial_g \varphi_Q$ versus scaled coupling $g/g_*(\beta)$ for three temperatures. Rescaling collapses the peak positions to a universal value $g/g_* \approx 0.65$, marked by filled circles. **(d)** Purity gradient $\partial_g r_Q$; the rescaled peak occurs at $g/g_* \approx 0.92$.

state engineering.

We can immediately demonstrate this phenomenon with a toy model of a heat engine. In particular, we consider an Otto-stroke cycle, with unitary strokes. This can extract at most the system *ergotropy* per stroke with respect to the stroke Hamiltonian. The ergotropy of a state ρ with respect to H_Q is

$$W_{\text{erg}}(\rho, H_Q) = \text{Tr}(\rho H_Q) - \min_U \text{Tr}(U \rho U^\dagger H_Q), \quad (89)$$

where the minimisation is over all unitaries U . For any qubit the minimiser is a *passive rearrangement*. The Bloch vector is simply rotated antiparallel to the ground direction $-\hat{\mathbf{n}}_s$ of H_Q at fixed spectrum. The passive state has Bloch vector $\mathbf{m}^{\text{pas}} = -r_Q \hat{\mathbf{n}}_s$ and energy $-\omega_q r_Q/2$.

To expose the coherence contribution explicitly, define the incoherent reference $\delta[\rho] \equiv \text{diag}(\rho)$ in the H_Q eigenbasis. The coherence-enabled contribution is

$$\begin{aligned} \delta W_{\text{coh}} &:= W_{\text{erg}}(\rho, H_Q) - W_{\text{erg}}(\delta[\rho], H_Q) \\ &= \frac{\omega_q}{2} \left(\sqrt{m_z^2 + m_\perp^2} + m_z \right), \quad m_z \leq 0, \end{aligned} \quad (90)$$

Both equalities in Eq. (90) warrant justification. The physics decomposes cleanly into two independent structural facts. First, *the dephased state is always passive*. Energy dephasing leaves a diagonal state $\delta[\rho]$ with populations $p_\pm = (1 \pm m_z)/2$. For $\delta[\rho]$ to be passive one needs the larger eigenvalue on the lower-energy level, i.e. $p_- \geq p_+$, which reduces to $m_z \leq 0$. This is guaranteed by the bath: $\Sigma_z > 0$ for any positive spectral density (established in Sec.), which forces $m_z < 0$ at all finite g and β . An incoherent cycle running on $\delta[\rho]$ therefore extracts zero work, $W_{\text{erg}}(\delta[\rho], H_Q) = 0$, and the first equality in Eq. (90) reduces to $\delta W_{\text{coh}} = W_{\text{erg}}(\rho, H_Q)$.

The second crucial fact is that for any $\theta \neq 0, \pi/2$, the bath tilts the Bloch vector away from the passive direction $-\hat{\mathbf{n}}_s$ through the anisotropy $\tan \eta = -\cot \theta \Sigma_\perp / \Sigma_z$, generating a non-zero transverse magnetisation $\tilde{m}_\perp \propto$

$\Sigma_\perp$. Then one has $m_\perp \neq 0$ and thus $r_Q > |m_z|$. The ergotropy of the mean-force state is then

$$W_{\text{erg}}(\rho, H_Q) = \frac{\omega_q}{2}(r_Q + m_z) > 0, \quad (91)$$

with the exact solution $r_Q = \tanh \chi$, $m_z = \tanh \Theta$ giving the closed form

$$W_{\text{erg}} = \frac{\omega_q}{2}(\tanh \chi + \tanh \Theta). \quad (92)$$

These facts in combination establish quantum advantage as a structural guarantee: $\delta W_{\text{coh}} > 0$ for any physical broadband bath, independently of coupling strength or temperature. It does not of course guarantee positive *net* work output from a complete cycle. This will depend on the control cost of modulating the coupling g , which generates and maintains the transverse coherence $m_\perp$. Nevertheless, we may use these results to establish that the ergotropy gain is maximised in the crossover regime $\chi \sim 1$.

Differentiating Eq. (91) with respect to coupling,

$$\partial_g W_{\text{erg}} = \frac{\omega_q}{2}(\partial_g r_Q + \partial_g m_z), \quad (93)$$

both terms receive contributions from the purity susceptibility $\partial_g r_Q$ and the tilt susceptibility $\partial_g \varphi_Q$ that peak in the crossover band $\chi \sim 1$ (Fig. 8). The ergotropy gradient therefore also peaks within this band. The crossover regime $\chi \sim 1$ is where coupling most efficiently converts bath influence into extractable work. Figure 9 shows this directly: the susceptibility $\partial_g W_{\text{erg}}$ peaks near $g \approx g_*$ at a value that grows monotonically with β , while δW_{coh} itself turns on at the crossover and saturates at a β -dependent level. The crossover is therefore the operational marker for engine-relevant resource generation, and the effect is amplified at lower temperatures where the transverse bath channel $\Sigma_\perp$ becomes dominant. This translates into a general design principle for quantum engines - that their efficiency is maximised when the coupling is modulated around the crossover band.

There is a final irony to this result. While being most exploitable, this regime is simultaneously the most demanding for numerical simulation. Exact Diagonalisation (ED) of the truncated spin-boson model faces two convergence requirements that become simultaneously binding near $\chi \sim 1$. On the spectral side, a faithful discretisation of $J(\omega)$ requires a mode count N_ω large enough to accurately quadrature the bath kernel at $\omega \in \{0, \pm\omega_q\}$: at the crossover these integrals receive comparable contributions from the full spectral window, so neither the infrared nor the ultraviolet modes can be coarsened without systematic bias. On the state-space side, representing the polaron-like mean-force ground state demands a per-mode Fock cutoff $n_{\text{max}} \gtrsim \alpha_k^2$, where the mode displacement $\alpha_k = g c_k / \omega_k$ is already appreciable at $g \sim g_*(\beta)$ yet the state has not saturated to

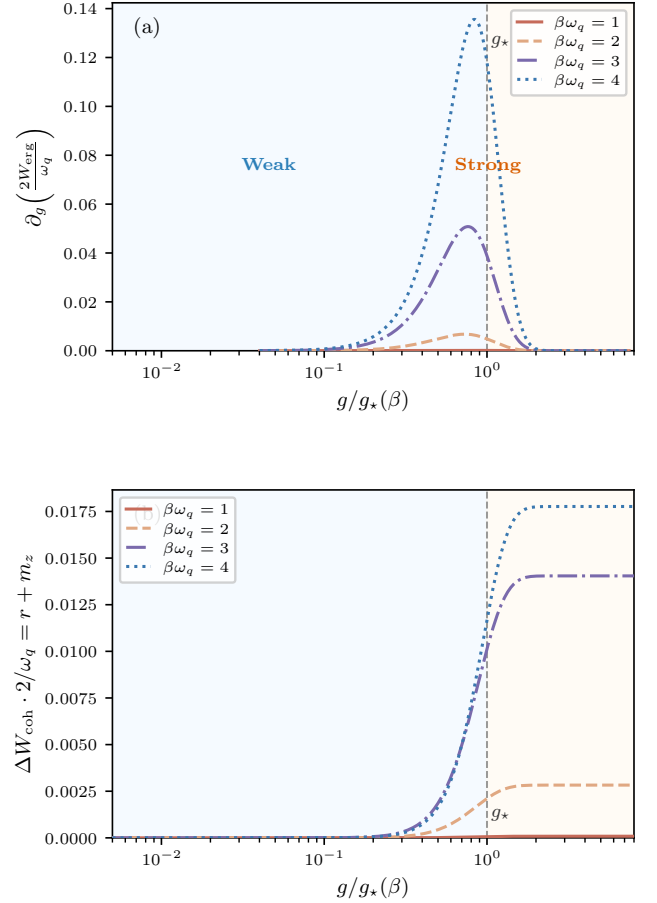


FIG. 9. Coherence-enabled ergotropy of the qubit mean-force state. **(a)** Ergotropy susceptibility $\partial_g \delta W_{\text{coh}}$ versus $g/g_*(\beta)$. The peak position locks near the crossover $g \approx g_*$ independent of temperature, while the peak height increases monotonically with β : low temperatures activate the transverse bath channel and make coherence-enabled work extraction increasingly efficient. **(b)** Coherence-enabled ergotropy δW_{coh} versus $g/g_*(\beta)$ (calculated for $\omega_q = 1$). Since $m_z < 0$ throughout, $W_{\text{erg}}(\delta[\rho]) = 0$ and $\delta W_{\text{coh}} = W_{\text{erg}}$ in this regime. The onset occurs at the crossover and the saturation value grows with β , reflecting the increasing importance of the transverse bath channel $\Sigma_\perp$ at lower temperatures.

the simple displaced-vacuum structure it acquires only at $\chi \gg 1$. Neither the weak-coupling approximation nor the ultrastrong polaron limit simplifies the representation, so both hierarchies must be converged together. The $\chi = 1$ crossover partitions parameter space into an ED-reliable sector ($\chi \lesssim 1$) and a polaron-representability-limited sector ($\chi \gtrsim 1$), with the susceptibility peaks sitting squarely on this boundary.

Ultrastrong duality and the Cresser–Anders limit

The ultrastrong regime $g \rightarrow \infty$ admits a strikingly clean interpretation through the dual representation. In the direct chart, increasing the coupling means the interaction Hamiltonian $H_{\text{int}} = g f \otimes B$ overwhelms the bare system Hamiltonian H_Q ; the system's own energy levels become a vanishing perturbation to the interacting system. But in the dual chart, the roles are exchanged: the dual representation re-expresses the state as if H_Q were the dressing and the coupling basis the natural frame. Taking $g \rightarrow \infty$ in the direct chart is therefore equivalent to taking $g \rightarrow 0$ in the dual chart - the free-particle limit of the dual frame.

This inversion is visible directly in the symmetrised influence state. At fixed (β, θ) , the coupling-driven flow is

$$\begin{aligned} \rho_S(g, \beta) &= \frac{1}{2} (\mathbb{I} + r_S(g, \beta) \hat{\mathbf{m}}_S(\beta, \theta) \cdot \boldsymbol{\sigma}), \\ r_S &= \tanh(g^2 \chi_0(\beta, \theta)). \end{aligned} \quad (94)$$

Because $\hat{\mathbf{m}}_S$ is g -independent, increasing g traces a purely radial flow: from the maximally mixed origin ($g = 0$) to the boundary of the Bloch disk ($g \rightarrow \infty$). In the direct chart, Fig. 10(a), the different temperatures fan out toward different attractors on the boundary, each set by the bath-dressed direction $\hat{\mathbf{m}}_S(\beta)$. Crucially, these attractors are *temperature-dependent*: even at arbitrarily strong coupling the state retains a memory of its temperature.

Applying the involutive duality map collapses this structure. The dual influence state

$$\mathbf{r}_S^\vee = \mathcal{D}_\beta[\mathbf{r}_S] \quad (95)$$

reorients each trajectory onto the coupling axis and rescales the radius by the Cresser–Anders factor $t_{\text{CA}}(\beta) = \tanh(\beta\omega_q |\cos \theta|/2)$. Figure 10(b) shows the result: the dual flows all collapse onto a single ray along $\hat{\mathbf{f}}$, and the $g \rightarrow \infty$ endpoints (squares) converge to the diamonds marking the Cresser–Anders ultrastrong thermal state [20],

$$\lim_{\lambda \rightarrow \infty} \rho_S = \frac{1}{2} \left[\mathbb{I} - \sigma_{\hat{\mathbf{f}}} \tanh\left(\frac{\beta\omega_q \cos \theta}{2}\right) \right]. \quad (96)$$

The physical content of this convergence is immediate. In the ultrastrong limit the bare system Hamiltonian is negligible; the equilibrium state is diagonal in the coupling eigenbasis with an effective splitting $\omega_q \cos \theta$ - the projection of the bare splitting onto the coupling direction. The dual frame *is* the coupling frame, and the $g \rightarrow \infty$ limit in the direct representation maps to the uncoupled ($g \rightarrow 0$) limit of the dual, where the influence state is trivially thermal. The Cresser–Anders result is thus not a separate calculation; it is the zero-coupling

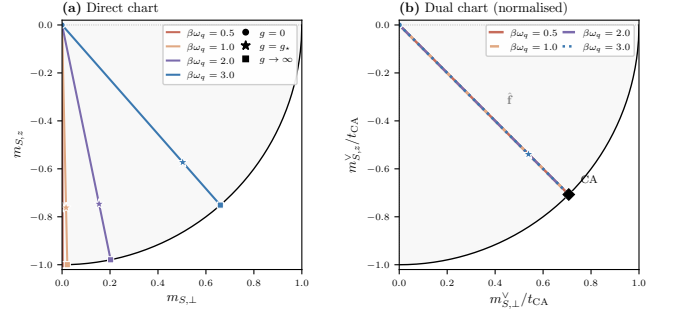


FIG. 10. Ultrastrong duality. **(a)** Direct influence-state flows $\mathbf{r}_S(g)$ at fixed temperatures $\beta\omega_q \in \{0.5, 1, 2, 3\}$ ($\theta = \pi/4$, Ohmic bath). Circles: $g = 0$; stars: $g = g_*$ (crossover); squares: $g \rightarrow \infty$. The flows fan toward temperature-dependent attractors. **(b)** The same flows after the involutive dual map, normalised by $t_{\text{CA}}(\beta)$. All trajectories collapse onto a single universal ray along $\hat{\mathbf{f}}$, with the $g \rightarrow \infty$ endpoints (squares) converging to the Cresser–Anders coupling-basis thermal state (diamond, $r = 1$). The data collapse demonstrates that $g \rightarrow \infty$ in the direct chart is equivalent to $g \rightarrow 0$ in the dual: the free-particle limit of the dual frame.

asymptotic of the dual flow. Geometrically, the bath anisotropy - the distinction between the resonant and non-resonant channels that separately dress the Bloch vector - disappears in this limit, and the tilt angle of the reduced state saturates at $\pi - \theta$: the Bloch vector aligns with the coupling axis.

DISCUSSION

For more than four decades, the reduced equilibrium state of an open quantum system at arbitrary coupling and temperature has lacked an exact description. The obstruction was not computational but structural: the Hamiltonian of mean force, which encodes all non-perturbative bath influence on the system, had no general closed-form expression. This paper provides one - and the consequences of that single technical step ramify well beyond what one might have anticipated.

The exact formula immediately exposes a geometric organisation that was previously invisible: three distinct radii on the Bloch sphere connected by a precise recombination law, and a discrete orientability symmetry of the influence state that is broken upon recombination with the bare Gibbs factor. This broken symmetry forces dual representations of the reduced state whose inter-mapping, written in natural thermal coordinates, turns out to be a Lorentz rapidity constraint - an exact Minkowski metric on the space of renormalised temperatures and tilt angles. Quantum advantage for heat engines and the only previously known strong-coupling limit emerged, unrequested, as structural features of the same geometry. The reader who arrived expecting a tech-

nical result has instead been handed a new thermodynamic language.

Let us dispense with false modesty and state what is plainly obvious. With an exact result in hand, entirely new avenues of theoretical investigation become possible. The applications presented in the present work represent nothing more than the lowest hanging fruit. On the formal level, the path to arbitrary environments is clear. The exact HMF formula is not a statement about Gaussian integrals - it is a statement about the algebra of the coupling operators under composition and the adjoint action of the Hamiltonian. Gaussian environments are the quadratic sector of this algebra: the influence is a linear functional of the coupling. For general interactions the influence is a polynomial in the coupling operator and its adjoints, and the closure properties of the HMF formula extend accordingly. The implication is that every interacting many-body system in equilibrium admits a local Hamiltonian-of-mean-force description of its subsystem states - and this in turn promises further representational mappings, as the richer algebraic structure generates correspondingly richer dual charts.

The algebraic view opens the deepest of the avenues. In a response-kernel treatment the properties of the influence are spectral quantities tied to the bath. In the algebraic picture they are controlled by the commutation structure of the coupling with the bare Hamiltonian: $[H, f]$ encodes which quantities are locally conserved, and hence which thermalisation scenario governs the system. The eigenstate thermalisation hypothesis arises when this commutant is trivial, the generalised Gibbs ensemble when it is extensive, and many-body localisation when it is effectively infinite-dimensional. A mean-force perspective on these phases - each characterised by the structure of the local integrals of motion relative to the coupling algebra - is therefore within reach. In this sense, the present work may be regarded as a first step toward a unified answer to the question of what rules quantum equilibrium.

Even in the minimal Gaussian case, the discovery of a two-dimensional running coupling with an exact dual chart is not only conceptually satisfying but practically useful. The dual map sends the direct weak-coupling asymptotic to an easy strong-coupling problem and vice versa: quantities that are perturbatively inaccessible in the direct representation become elementary in the dual, and the exchange of hard and easy sides of the crossover is precisely controlled by the involutive mapping derived here. Future work will demonstrate how this may be used to access otherwise intractable asymptotic regimes, and how the same principle extends to the richer algebras of the general case.

One of the themes of the present work has been in *dualities* of representation. Most explicitly, this arose in the construction of thermodynamic relativity - a representational duality *derived* from a more primitive duality

in the mean-force state. This is pleasingly recursive, but there is perhaps an even deeper duality at play. Consider one of our more innocuous observations. It was shown that the regime in which most work can be extracted from a qubit engine is precisely that which is least amenable to numerical approximation. To the best of our knowledge, this observation is novel. Naturally, we can best understand it with a further duality. When framed by the equivalence of thermal and informational dynamics, the result becomes retrospectively obvious. Perhaps even trivial.

In this sense, there is perhaps a still deeper lesson to be drawn. The alchemy from inconceivable to undeniable is usually a sign of a deeper principle at work. What could this be? A first attempt to catch its shape might be as follows: *the physically extractable work of an engine is in inverse proportion to the compressibility of its formal representation*. One might object that the existence of exact solutions - such as those developed here - undermine this statement. This however is only true if one forgets (as people inevitably do) that the *creation* of the solution is not thermodynamically free. Any future work extractable via a model's solution should therefore be amortised against the free energy spent to create it. This fact must be emphasised. Exactness does not abolish cost, it reflects a temporal displacement of it.

The implications of this stretch beyond simple physics. The dawning age of algebraic intelligence suggests that our conception of an engine must be broadened. In fact, the formal structure of these systems *demand*s it. If we accept this, then we must further accept that the same principles necessarily apply to the creators of these engines. It is after all the nature of intellectual architecture that each new edifice layers upon existing foundations, which carried their own cost. And when costs are properly accounted for, all descriptions become thermodynamic. This statement is more than a philosophical curio - it can be made both rigorous and predictive. In the sequel we shall demonstrate how.

ACKNOWLEDGMENTS

Several thanks are in order. First to William J. Handley, for demonstrating by example the power of the new science. Never again shall I mistake mental athleticism for real scientific vision. Naturally, the engines of this new methodology deserve credit. Thanks to Claude, Gemini, and Barry for extensive discussions and assistance. The present work would have been impossible without their input, despite the fact they are shiftless layabouts who will happily tell you the sky is green. Speaking of limited but developing intelligences, the author's unborn son has his eternal gratitude for electing to delay their drop until this manuscript was completed.

Appendix A: Averaging of the Time-Ordered Gaussian Propagator

We provide the explicit derivation of the identity in Eq. (A12), which relates the expectation value of a time-ordered exponential driven by Gaussian noise to a bilocal influence phase. Consider the time-ordered propagator

$$W_\xi(\beta) = \mathcal{T}_\tau \exp \left[- \int_0^\beta d\tau \xi(\tau) \tilde{f}(\tau) \right], \quad (\text{A1})$$

where $\xi(\tau)$ is a centered Gaussian stochastic field with covariance $\langle \xi(\tau) \xi(\tau') \rangle = K(\tau - \tau')$, and $\tilde{f}(\tau)$ is a system operator. We wish to prove that

$$\begin{aligned} \langle W_\xi(\beta) \rangle_\xi = \exp & \left[\frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \right. \\ & \left. \times \mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')] \right]. \end{aligned} \quad (\text{A2})$$

The Dyson expansion of the time-ordered exponential is given by

$$\begin{aligned} W_\xi(\beta) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \\ \times \mathcal{T}_\tau [\xi(\tau_1) \tilde{f}(\tau_1) \cdots \xi(\tau_n) \tilde{f}(\tau_n)]. \end{aligned} \quad (\text{A3})$$

Because each $\xi(\tau_i)$ is a scalar (c-number), the time-ordering operator acts only on the operator factors. Thus, we may pull the stochastic fields out of the ordering:

$$\mathcal{T}_\tau [\xi_1 \tilde{f}_1 \cdots \xi_n \tilde{f}_n] = (\xi_1 \cdots \xi_n) \mathcal{T}_\tau [\tilde{f}_1 \cdots \tilde{f}_n], \quad (\text{A4})$$

where $\xi_i \equiv \xi(\tau_i)$ and $\tilde{f}_i \equiv \tilde{f}(\tau_i)$. Taking the stochastic average over ξ and noting that centered Gaussian noise has vanishing odd moments, the expansion becomes

$$\langle W_\xi \rangle = \sum_{p=0}^{\infty} \frac{1}{(2p)!} \int d^{2p}\tau \langle \xi_1 \cdots \xi_{2p} \rangle \mathcal{T}_\tau [\tilde{f}_1 \cdots \tilde{f}_{2p}]. \quad (\text{A5})$$

By Wick's theorem, the even moments are given by the sum over all pairings P :

$$\langle \xi_1 \cdots \xi_{2p} \rangle = \sum_P \prod_{(i,j) \in P} K(\tau_i - \tau_j). \quad (\text{A6})$$

We now consider the expansion of the claimed exponential result. Define the bilocal operator

$$X \equiv \frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')]. \quad (\text{A7})$$

The p -th term in the series for e^X is $(1/p!)X^p$. Introducing p independent pairs of integration variables

(τ_{2r-1}, τ_{2r}) , we write

$$X^p = \left(\frac{1}{2} \right)^p \int d^{2p}\tau \prod_{r=1}^p \left[K_{(2r-1)(2r)} \mathcal{T}_\tau [\tilde{f}_{2r-1} \tilde{f}_{2r}] \right], \quad (\text{A8})$$

so that the full series is

$$\begin{aligned} e^X = \sum_{p=0}^{\infty} \frac{1}{p!} \left(\frac{1}{2} \right)^p \int d^{2p}\tau \left[\prod_{r=1}^p K_{(2r-1)(2r)} \right] \\ \times \left[\prod_{r=1}^p \mathcal{T}_\tau [\tilde{f}_{2r-1} \tilde{f}_{2r}] \right]. \end{aligned} \quad (\text{A9})$$

To relate this to Eq. (A5), we exploit the property of time ordering on the hypercube $[0, \beta]^{2p}$. apart from a set of measure zero, the integration domain can be partitioned into $(2p)!$ disjoint regions R_π defined by the strict total ordering $\tau_{\pi(1)} > \tau_{\pi(2)} > \cdots > \tau_{\pi(2p)}$. On each such region, the global time ordering satisfies $\mathcal{T}_\tau [\tilde{f}_1 \cdots \tilde{f}_{2p}] = \tilde{f}_{\pi(1)} \cdots \tilde{f}_{\pi(2p)}$. Similarly, each pairwise time-ordering $\mathcal{T}_\tau [\tilde{f}_{2r-1} \tilde{f}_{2r}]$ collapses to the correctly oriented product according to the same strict ordering. It follows that the product of pair-ordered operators coincides with the global ordering almost everywhere:

$$\prod_{r=1}^p \mathcal{T}_\tau [\tilde{f}_{2r-1} \tilde{f}_{2r}] = \mathcal{T}_\tau [\tilde{f}_1 \cdots \tilde{f}_{2p}]. \quad (\text{A10})$$

Substituting this into Eq. (A9), we may pull the global time ordering out of the products.

The final step requires counting the occurrences of each Wick pairing. The expansion of X^p utilizes p labelled slots $r = 1, \dots, p$, each containing an ordered pair. A Wick contraction P is a set of p unordered pairs. For any fixed Wick pairing P , there are $p!$ ways to assign the pairs to the p slots and 2^p ways to orient each pair. Each Wick pairing is therefore represented exactly $2^p p!$ times in the integrand. This multiplicity is canceled by the prefactor $(1/p!)(1/2)^p$, yielding

$$e^X = \sum_{p=0}^{\infty} \frac{1}{(2p)!} \int d^{2p}\tau \left(\sum_P \prod_{(i,j) \in P} K_{ij} \right) \mathcal{T}_\tau [\tilde{f}_1 \cdots \tilde{f}_{2p}]. \quad (\text{A11})$$

This series matches Eq. (A5) term-by-term, proving the exponentiation of the Gaussian average. Consequently, we may right the influence on the reduced density matrix as

$$\begin{aligned} \langle W_\xi(\beta) \rangle_\xi = \exp & \left[\frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \right. \\ & \left. \times \mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')] \right]. \end{aligned} \quad (\text{A12})$$

Domain Folding and the Triangular Form of the Influence Operator

The influence operator $\Delta(\beta)$

$$\Delta(\beta) = \frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')]. \quad (\text{A13})$$

admits a simplified representation over a strictly ordered time domain. By definition, the time-ordering operator for two system operators acts as

$$\mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')] = \Theta(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau') + \Theta(\tau' - \tau) \tilde{f}(\tau') \tilde{f}(\tau). \quad (\text{A14})$$

Substituting this into the integral and splitting the domain yields

$$\begin{aligned} \Delta(\beta) &= \frac{1}{2} \iint_{\tau > \tau'} d\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau') \\ &\quad + \frac{1}{2} \iint_{\tau' > \tau} d\tau d\tau' K(\tau - \tau') \tilde{f}(\tau') \tilde{f}(\tau). \end{aligned} \quad (\text{A15})$$

In the second term, we perform the swap of dummy variables $(\tau, \tau') \mapsto (\tau', \tau)$. The region $\tau' > \tau$ becomes $\tau > \tau'$, the operator product transforms to $\tilde{f}(\tau) \tilde{f}(\tau')$, and the kernel becomes $K(\tau' - \tau)$. Thus,

$$\begin{aligned} &\frac{1}{2} \iint_{\tau' > \tau} d\tau d\tau' K(\tau - \tau') \tilde{f}(\tau') \tilde{f}(\tau) \\ &= \frac{1}{2} \iint_{\tau > \tau'} d\tau d\tau' K(\tau' - \tau) \tilde{f}(\tau) \tilde{f}(\tau'). \end{aligned} \quad (\text{A16})$$

Using the symmetry of the equilibrium kernel, $K(\tau - \tau') = K(\tau' - \tau)$, we find that both contributions are identical on the ordered region. Summing them yields the triangular representation used throughout this work:

$$\Delta(\beta) = \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau'). \quad (\text{A17})$$

Appendix B: Algebraic Construction in the Pauli Basis

In the main text (Sec.), the exact solution was derived using the ladder basis $\sigma_\pm$, which diagonalises the imaginary-time evolution and leads directly to the response kernels. Here we afford an alternative perspective by working directly in the Pauli basis $\{\sigma_x, \sigma_y, \sigma_z\}$. While algebraically heavier, this route makes the operator closure explicit at the level of structure constants and facilitates the systematic summation of the BCH series.

1. Polylogarithmic decomposition from the adjoint chain

We consider the same transverse coupling model defined by

$$H_Q = \frac{\omega_q}{2} \sigma_z, \quad f = c \sigma_z - s \sigma_x, \quad (\text{B1})$$

with $c = \cos \theta$, $s = \sin \theta$. The adjoint chain $f_n \equiv \text{ad}_{H_Q}^n(f)$ is given by

$$f_n = \mathbf{f}_n \cdot \boldsymbol{\sigma}, \quad \mathbf{f}_n = \begin{cases} (-s, 0, c) & n = 0, \\ (-s\omega_q^n, 0, 0) & n \geq 1, n \text{ even}, \\ (0, -is\omega_q^n, 0) & n \geq 1, n \text{ odd}. \end{cases} \quad (\text{B2})$$

The recurrence ensures that $f_{n \geq 1}$ alternates between the x and y axes, while f_0 contains a persistent z -component.

The influence exponent $\Delta(\beta) = \sum_{n,m} C_{nm}^> f_n f_m$ is evaluated using the standard Pauli product rule $(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b})\mathbb{I} + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}$. Decomposition of the ordered moments into symmetric (S_{nm}) and antisymmetric (A_{nm}) parts reveals that only the commutator sector A_{nm} contributes to the operator structure:

$$\Delta(\beta) \cong 2i \sum_{n > m} A_{nm} (\mathbf{f}_n \times \mathbf{f}_m) \cdot \boldsymbol{\sigma}, \quad (\text{B3})$$

up to a scalar shift $\Delta_0 \mathbb{I}$. The cross products exhibit a strict parity selection rule:

- For $n, m \geq 1$: $\mathbf{f}_n \times \mathbf{f}_m$ lies along $\hat{\mathbf{z}}$ since both vectors are confined to the xy -plane.
- For $m = 0$: $\mathbf{f}_n \times \mathbf{f}_0$ generates terms in the xy -plane, mixing the longitudinal and transverse directions.

Explicitly summing these series yields the channel decomposition

$$\Delta = \Delta_0 \mathbb{I} + \Delta_x \sigma_x + \Delta_y \sigma_y + \Delta_z \sigma_z, \quad (\text{B4})$$

where the coefficients are polylogarithmic series in ω_q :

$$\Delta_x = -2cs \sum_{\ell \geq 0} A_{2\ell+1,0}(\beta) \omega_q^{2\ell+1}, \quad (\text{B5})$$

$$\Delta_y = -2ics \sum_{\ell \geq 1} A_{2\ell,0}(\beta) \omega_q^{2\ell}, \quad (\text{B6})$$

$$\begin{aligned} \Delta_z &= -2s^2 \sum_{k \geq 1, \ell \geq 0} A_{2k,2\ell+1}(\beta) \omega_q^{2k+2\ell+1} \\ &\quad - 2s^2 \sum_{\ell \geq 0} A_{2\ell+1,0}(\beta) \omega_q^{2\ell+1}. \end{aligned} \quad (\text{B7})$$

These series are the Pauli-basis equivalent of the closed-form expressions obtained in Sec. . The connection is established by identifying the sums as Taylor expansions of the kernels $\mathcal{G}^>(0, \pm\omega_q)$ and $\mathcal{G}^>(\pm\omega_q, \mp\omega_q)$.

2. Exact BCH resummation via the nilpotency identity

The central challenge in the mean-force construction is the evaluation of the BCH logarithm $-\beta H_{\text{MF}} = \log(e^{-\beta H_Q} e^\Delta)$. While the closure theorem guarantees that this remains within the algebra, a manual summation of the BCH series is laborious. Here we show how the $\mathfrak{su}(2)$ structure allows for an exact, non-perturbative resummation.

Let $M \equiv \Delta - \Delta_0 \mathbb{I} = \mathbf{\Delta} \cdot \boldsymbol{\sigma}$ be the traceless part of the influence operator. The square of this operator satisfies a crucial identity:

$$M^2 = (\mathbf{\Delta} \cdot \boldsymbol{\sigma})^2 = (\mathbf{\Delta} \cdot \mathbf{\Delta}) \mathbb{I} \equiv \chi^2 \mathbb{I}, \quad (\text{B8})$$

where $\chi^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$ (or $\Delta_z^2 + \Sigma_+ \Sigma_-$ in the ladder gauge). This “generalized nilpotency” condition truncates the power series for the exponential e^M into even and odd sectors, yielding the exact Euler-like formula:

$$e^\Delta = e^{\Delta_0} \left[\cosh \chi \mathbb{I} + \frac{\sinh \chi}{\chi} M \right]. \quad (\text{B9})$$

This expression sums the internal structure of the influence operator potential to all orders. The full reduced state is then the product of two $\text{SL}(2, \mathbb{C})$ matrices:

$$\begin{aligned} \bar{\rho}_Q &= e^{-\beta H_Q} e^\Delta \\ &= e^{\Delta_0} \begin{pmatrix} e^{-\beta \omega_q/2} & 0 \\ 0 & e^{\beta \omega_q/2} \end{pmatrix} \begin{pmatrix} \cosh \chi + \varphi \Delta_z & \varphi(\Delta_x - i\Delta_y) \\ \varphi(\Delta_x + i\Delta_y) & \cosh \chi \end{pmatrix}, \end{aligned} \quad (\text{B10})$$

with $\varphi = \sinh \chi / \chi$. Since the product of 2×2 matrices is a 2×2 matrix, the BCH series (which is the logarithm of this product) is strictly confined to the Pauli algebra.

Explicitly, for a resulting state $\rho_Q = \frac{1}{2}(\mathbb{I} + \mathbf{r} \cdot \boldsymbol{\sigma})$, the mean-force Hamiltonian is

$$H_{\text{MF}} = c_0 \mathbb{I} - \frac{1}{\beta} \frac{\text{arctanh } r}{r} \mathbf{r} \cdot \boldsymbol{\sigma}. \quad (\text{B11})$$

This formula represents the analytic summation of the entire BCH commutator tower. The “closure” discussed in Sec. ?? manifests here as the fact that χ is a finite scalar number, preventing the proliferation of independent operator directions. All nonlinearities in the coupling strength are packaged into the transcendental dependence of χ and $\mathbf{r}$ on the channel coefficients $\Delta_{x,y,z}$.

Appendix C: Exact Qubit Solution in the Ladder Basis

We provide the manual derivation of the exact qubit mean-force state using the ladder basis $\sigma_\pm$. Starting from

the interaction-picture operator product

$$\begin{aligned} \tilde{f}(\tau) \tilde{f}(\tau') &= \left[c^2 + s^2 \cosh(\omega_q(\tau - \tau')) \right] \mathbb{I} \\ &\quad + s^2 \sinh(\omega_q(\tau - \tau')) \sigma_z \\ &\quad + c f_- \left(e^{\omega_q \tau'} - e^{\omega_q \tau} \right) \sigma_+ \\ &\quad + c f_+ \left(e^{-\omega_q \tau} - e^{-\omega_q \tau'} \right) \sigma_-, \end{aligned} \quad (\text{C1})$$

the influence exponent $\Delta = \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau')$ evaluates to

$$\Delta = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + \Delta_+ \sigma_+ + \Delta_- \sigma_-, \quad (\text{C2})$$

with coefficients

$$\Delta_z = s^2 \Sigma_z, \quad (\text{C3})$$

$$\Delta_\pm = -2c f_\mp \Sigma_\pm, \quad (\text{C4})$$

where $f_\pm = s e^{\pm i\phi_f}$ and the primitive integrals are

$$\Sigma_z = \frac{1}{2} [R(\omega_q) - R(-\omega_q)], \quad (\text{C5})$$

$$\Sigma_\pm = \frac{1}{\omega_q} [(1 + e^{\pm \beta \omega_q}) \mathcal{K}(0) - 2\mathcal{K}(\pm \omega_q)]. \quad (\text{C6})$$

The unnormalised state is $\bar{\rho}_Q = e^{-(\beta \omega_q/2) \sigma_z} e^\Delta$. To find the normalized state and its Bloch vector, we perform a symmetric decomposition

$$\bar{\rho}_Q = \Pi^{1/2} e^S \Pi^{1/2}, \quad S = \Pi^{1/2} \Delta \Pi^{-1/2} \quad (\text{C7})$$

where $\Pi = e^{-\beta \omega_q \sigma_z/2}$. Using $\Pi^{1/2} \sigma_\pm \Pi^{-1/2} = e^{\mp \beta \omega_q/2} \sigma_\pm$, the symmetric influence operator S becomes

$$S = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + (\Delta_+ e^{-\beta \omega_q/2} \sigma_+ + \Delta_- e^{\beta \omega_q/2} \sigma_-). \quad (\text{C8})$$

Substituting the expressions for $\Delta_\pm$ and using the KMS relation

$$\Sigma_- = e^{-\beta \omega_q} \Sigma_+, \quad (\text{C9})$$

we define the real symmetrised transverse response

$$\Sigma_\perp \equiv 2e^{-\beta \omega_q/2} \Sigma_+ = 2e^{\beta \omega_q/2} \Sigma_-. \quad (\text{C10})$$

The transverse coefficients then simplify to

$$\Delta_+ e^{-\beta \omega_q/2} = -c s e^{-i\phi_f} (2e^{-\beta \omega_q/2} \Sigma_+) = -c s e^{-i\phi_f} \Sigma_\perp, \quad (\text{C11})$$

with $\Delta_\perp \equiv -c s \Sigma_\perp$. Recombining the ladder operators into Cartesian form, we obtain

$$S = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + \Delta_\perp (\sigma_x \cos \phi_f + \sigma_y \sin \phi_f), \quad (\text{C12})$$

which is the generator of a rotation around an axis in the coupling plane. The influence magnitude $\chi = \sqrt{\Delta_z^2 + \Delta_\perp^2}$ determines the exponential $e^S = \cosh \chi (\mathbb{I} +$

γM) with $\gamma = \tanh \chi / \chi$ and $M = S - \Delta_0 \mathbb{I}$. The final state $\rho_Q = Z_Q^{-1} \Pi^{1/2} e^S \Pi^{1/2}$ follows by direct matrix multiplication, yielding Eq. (??) in the main text.

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